

The Impact of Refugee Arrivals on Food Security, Income, and Crop Prices in Rural Uganda

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Abstract

This study estimates the impacts of refugee inflows on household food security and agricultural production in Uganda. Using a representative panel of data on refugees and hosts from 2017–2021, I find that refugee arrivals cause statistically significant food security deterioration. Scaled to average annual arrivals (9,182), host households experience a 9.1-point (0.55 SD) increase in the Coping Strategies Index and established refugees an 11.4-point (0.44 SD) increase. These impacts push the average household into the bottom quartile of baseline food security, driven primarily by net consumer host households and net producer refugee households. Despite adverse consumption effects, agricultural market participation expands: host and established refugee households are 26.6 and 11.2 percentage points more likely to report agricultural earnings, respectively. For host households, this market entry is coupled with a significant accumulation of livestock, evidenced by a 68 percent increase in total holdings over 12 months. Refugee households, however, exhibit no such accumulation. The decline in food security, despite market extensification, is consistent with delayed, arrival-driven price inflation for both aid and non-aid food commodities. These price effects erode purchasing power, disproportionately harming net food consumers and households with limited production capacity or aid. These findings underscore the importance of agricultural support and targeted in-kind aid in rural refugee-hosting regions.

JEL Classification: O15, Q18, F22, R23

Keywords: Refugees, Food Security, Crop Prices, Displacement, Uganda

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1 Introduction

The scale of global displacement reached unprecedented levels at the end of 2023, affecting nearly 120 million people worldwide, equivalent to over one in every seventy individuals globally. This number has grown consecutively for the past twelve years, and is projected to have increased again in 2024 (UNHCR, 2024). While the hosting of displaced persons often dominates political discourse in high-income Western nations, the reality is that the vast majority of displaced people (approximately 75%) find refuge in low- and middle-income countries, typically within their own regions. These host countries frequently face high poverty rates and constrained capacity to support new arrivals adequately. The arrival of a large displaced population into a settlement can alter local markets, creating significant increases in demand for essential goods, particularly food. These issues raise critical questions about the impacts of large refugee inflows. Specifically, this paper estimates the effects of large-scale refugee arrivals on settlement communities' food security and agricultural production, alongside the corresponding price mechanism.

In East and Central Africa, current instability is causing large-scale displacement. The UNHCR describes the situation in South Sudan as "Africa's largest refugee crisis", which is compounded by chronic underfunding of aid efforts (UNHCR, 2025). The situation in the Democratic Republic of the Congo (DRC) is similarly described as one of the most protracted and complex humanitarian disasters. Together, these crises have forcibly displaced more than 20 million people. Located between these immense conflict zones, Uganda occupies a distinct position as a major recipient of refugees.

Uganda, Africa's largest and oldest refugee host nation, is widely regarded as a "model" host due to its unique and progressive policies. Refugees in Uganda are granted access to social services and education, various forms of aid (cash transfers, food, seeds, and farm inputs) and notably, the allocation of plots of land to new arrivals for cultivation (World Bank Group, 2016). This combination creates a distinct system that directly supports refugees in the short run, with medium- and long-term integration based on increasing levels of self-sufficiency (Brown & Torre, 2024). Over time, in-kind food aid declines, by-design, leading refugees to rely on local markets for food and inputs. This dual-aid system may promote production and self-reliance via land access. However, these declines in aid paired with low land quality and economic insularity may generate localized demand pressures and food price inflation, leading to adverse consequences for both host and refugee households.

To identify these dynamics, I use the Resilience Index Measurement and Analysis (RIMA) survey that spans four waves, from 2017-2021 across 11 refugee-hosting districts. This represents the most recent and geographically comprehensive dataset for refugees in Uganda. This is combined with monthly settlement-level price and arrivals data. The empirical strategy uses an Arellano-Bond GMM estimator to identify price effects

and a two-way fixed effects model using a shift-share instrument to estimate the impact of arrivals on both host and established refugee household food security, agricultural production and earnings. Together, these estimations allow me to identify both how arrivals affect household welfare and a corresponding mechanism.

I find that refugee arrivals lead to a statistically significant deterioration in food security for both host and established refugee households. Scaling to the average district-level 12-month increase in arrivals (9,182), host households experience a 9.1-point increase in the Coping Strategies Index, and established refugees an 11.4-point increase. This pushes the average household into the bottom quartile of baseline food security for each group. I estimate that for every 1,000 additional district-level refugee arrivals within the past year, households experience an increase in CSI of 0.99 points (hosts) and 1.23 points (established refugees), on average. These results reflect a greater adoption of behaviors like food rationing, borrowing, or skipping meals. Despite these adverse consumption outcomes, I observe a large expansion in agricultural market participation. Scaled to average inflows, host and established refugee households are 26.6 and 11.2 percentage points more likely to report positive agricultural earnings, respectively. Further, the estimations reveal a stark divergence in how these groups manage baseline capital and assets, specifically livestock. Host households initially liquidate livestock in the short term—experiencing a 16.3 percent decrease at the 3-month mark—but use this liquidity to rapidly accumulate assets in the medium- to long-term, increasing total herd size by 68 percent at the 12-month mark, when scaled to average inflows. Established refugees see no significant changes in livestock ownership across any time horizon, highlighting severe initial capital constraints.

I argue that the primary mechanism driving these diverging outcomes—worsened food security alongside increased market participation—is localized price inflation resulting from refugee arrivals. Although the price data overlaps with the household data for one year (2021), the level of refugees during this period is similar. Therefore, this analysis provides evidence of the relationship between prices and arrivals, an important channel for household outcomes. The price analysis demonstrates that arrivals lead to delayed but significant inflation for both aid and non-aid food commodities, with significant price effects emerging in the 6–12 month post-arrival period. This timing is consistent with aid policy that gradually reduces in-kind assistance, as well as the delayed impacts on food security and earnings for the household. These price effects erode household purchasing power, likely underlying the observed negative food security outcomes and increased market activity as households seek liquidity.

Notably, the outcomes for households are not uniform and depend on baseline market participation, initial capital, and the modality of aid received. Dividing the sample into "producers" and "consumers" based on initial market participation reveals diverging outcomes by household and subgroup. Host producer households see no food security impacts and are able to capitalize on these demand shocks using livestock as a liquid wealth buffer to enter markets and accumulate wealth, a strategy unavailable to refugees due to binding

capital and land constraints. Conversely, host consumer households, with lower levels of capital and aid assistance, experience worsened food security with no improvements in earnings or production due to their dependence on purchased food and inability to adjust to price inflation.

Among refugee households, the story diverges. Refugee consumers with higher access to inflation-resistant food aid maintain baseline food security and rely on this direct aid to buffer consumption and allow for some levels of market participation. Refugee producer households, however, experience increased food insecurity despite increases in overall yields. Fixed land allocations, low levels of capital and inflation-prone cash aid prevent capitalization on price effects and constrain consumption smoothing. Because a larger share of their aid is distributed as cash-based vouchers, local price inflation severely erodes their purchasing power relative to in-kind food relief. These results underscore the critical importance of direct aid assistance and targeted agricultural support for hosting communities and settlements.

By addressing these questions, this study contributes to a growing body of work on the economic impacts of refugee inflows. Alix-Garcia et al. (2018) find a beneficial effect of refugee inflows on host community consumption in Kakuma, Kenya. The divergence from the adverse food security impacts found in this paper is largely driven by differing refugee policies and heterogeneous host characteristics found in Uganda. Kakuma operates as a strict encampment where local pastoralists were able to capitalize on local demand shocks, a response mirrored in the host producer findings in our heterogeneity analysis. Conversely, Uganda's agrarian integration model exposes net-consuming hosts to local food price inflation with limited ability to respond. On food prices, Alix-Garcia and Saah (2010) find that refugee arrivals in Tanzania generate significant short-term price increases for non-aid commodities, with the effects on aid goods more muted. They also find significant and positive effects for host household wealth in rural receiving areas. Alix-Garcia et al. (2012) extend this, showing that displacement from rural to urban settings lead to negative welfare outcomes for net consumers, while benefiting local providers of non-tradable goods. These results underscore the role of market structure, household characteristics, and aid access on welfare outcomes.

While food prices are a primary channel, there is a broader literature focusing on labor markets. Verme and Schuettler (2021) find in a meta-analysis of nearly 60 empirical studies that wage and employment effects of refugee arrivals are largely insignificant. Where effects exist, they most often demonstrate decreased employment and wages, particularly in low-to-middle income countries and informal sectors. These results suggest that the main welfare impacts of refugee arrivals in rural, agrarian settings may be transmitted less through labor markets and more through local prices as well as the quantity and types of aid present.

There is a small body of work on Uganda. Using a general equilibrium model to identify the impact of land distribution to new arrivals in Uganda, Zhu et al. (2024) find a positive welfare effect on host households due to income spillovers. They also find that the agricultural production sector is the largest beneficiary

of Uganda’s refugee aid program. Mastrorillo et al. (2024) examine the feasibility of Uganda’s self-reliance strategy using the RIMA survey’s first two waves (2017-2019). They find that both food and cash transfers improve food security, but their effectiveness depends on household resources and context: food transfers improve dietary outcomes, while cash transfers promote self-reliance only when refugees have sufficient land to cultivate. The study most related to this research, Kadigo and Maystadt (2023), estimate the effects of hosting refugees on material welfare in Uganda. The authors use LSMS data from 2010-2012 that weights the enumeration area’s distance to the nearest refugee settlement. They use a shift-share instrument to circumvent the endogeneity of the presence of refugee arrivals. Their findings demonstrate that the presence of refugees increases the likelihood of host households transitioning from subsistence to commercial agriculture, positively affecting earnings, consumption, and wages. These results appear to contrast with this paper’s findings of adverse food security effects while reinforcing the evidence that host households with stronger baseline production capacity capitalize on arrivals via earnings and livestock gains.

The divergence between the findings of Kadigo and Maystadt (2023) and Zhu et al. (2024) and those of this paper likely reflects the substantive changes in the Ugandan refugee context between the study periods. Zhu et al. (2024) largely uses data from early 2016, while Kadigo and Maystadt (2023)’s data originates from 2010 to 2012. Both periods predate the sudden and large-scale inflows from South Sudan and the Democratic Republic of the Congo (UNHCR, 2024) that is covered in the study period of this paper (2017–2021). The two studies also measure related but distinct treatment effects: Kadigo and Maystadt (2023) identify the average effect of proximity to a settlement on surrounding host communities, and Zhu et al. (2024) directly measure the effects of the land allocation in a generalized setting. I identify the marginal effect of sustained arrival flows on households already embedded in settlement communities.

This paper contributes to the literature on the economics of forced displacement in multiple dimensions. I use the largest and most geographically comprehensive dataset on Ugandan hosting areas to date: the RIMA survey from 2017-2021. The analysis combines estimates of household welfare outcomes with the identification of a price mechanism to provide an extensive overview of the economic impacts of arrivals on both host and established refugee communities simultaneously. The estimation also provides important insights into the temporal dynamics of these effects, documenting that negative impacts on food security coincide with the scheduled tapering of aid. Finally, I document the heterogeneous impacts of arrivals based on household market position and aid access. This study complements previous work, while offering a more complete empirical account of Uganda’s self-reliance model under the strain of long-term inflows, illustrating the roles that production capacity and aid modalities play in mediating these long-term impacts.

The remainder of this paper is organized as follows. Section 2 presents the theoretical framework and Section 3 describes the data. Sections 4 and 5 detail the empirical strategy and results for household food

security, earnings, and heterogeneous outcomes by market participation. Sections 6 and 7 present the price mechanism analysis. Section 8 concludes.

2 Theoretical Framework

This section builds a standard agricultural model (Singh et al., 1986) to analyze the impacts of refugee arrivals on food security in and around refugee settlements in Northern and Western Uganda. The arrival of displaced persons introduces the potential for increased aid and expansions in total production. This is combined with limited integration of local food markets (Titeca, 2020) leading to endogenous prices for food and labor. This is typical of food markets in rural areas with limited infrastructure and high transaction costs (Zhu et al., 2024).

The assumption of market insularity is supported by qualitative evidence collected by the author and others from multiple Ugandan settlements. This fieldwork—conducted in September 2025—consisted of interviews with four settlement commandants, four appointed refugee community leaders, and 25 refugee households, focusing on household production, food security outcomes, and local market characteristics. Several refugee residents cited the near impossibility of accessing larger markets. An elected community leader in the Imvepi settlement stated, "We would like to sell in bigger markets like Arua or Yumbe (both regional cities) to get better prices, but we don't have the capacity—we cannot afford the transportation". Another farmer in the Kyangwali settlement highlighted the impossibility of transportation, "There is no outside access to other markets (to buy or sell crops)". This sentiment was echoed in other areas, with refugees in Bidi-Bidi and Nakivale also noting the lack of outside market access due to prohibitive transport costs (Participants in the Refugee Settlement Study, 2025).

The features of these markets, combined with population influxes from new arrivals generate both supply and demand-side pressures within local food markets.

2.1 Model

Consider a local economy, defined as the designated refugee settlement and its surrounding population, where households act as both producers and consumers of a locally traded staple food crop. Production and consumption decisions also consider both cash transfer and direct food-aid.

2.1.1 Household Production

Host (H) and previously established refugee households (R) produce a composite food good, Q_a^i , while new arrivals (N), though potentially having access to land, are delayed in their production by one seasonal

cycle, leading to new arrival households not contributing to production in the current period. I assume a Cobb-Douglas household production technology for the staple food good:

$$Q_a^H = L_H^\alpha \bar{A}_H^\beta \quad (1)$$

$$Q_a^R = F_R^\alpha \bar{A}_R^\beta \quad (2)$$

where Q_a^i represents the agricultural output of household i . α and β are the labor and land output elasticities, and assumed to demonstrate constant returns to scale, $\alpha + \beta = 1$. L^H is the total labor input on host farms, composed of host family labor F_H and hired labor H_H , such that $L_H = F_H + H_H$. Established refugee households use only family labor F_R , and are assumed unable to hire labor, reflecting capital constraints. $\bar{A}_i$ represents the fixed plot of land available to household i . I assume that for refugee households, this land endowment will on average be smaller than their host counterpart, due to the settlement size constraints; $\bar{A}^H > \bar{A}^R > 0$. Refugees are officially allocated a fixed 30x30 meter plot for both housing and subsistence gardening. The land is often of poor quality, described as "rocky" and having poor drainage, especially in the northern settlements. As one refugee community leader noted, "At first, we were discouraged and didn't think anything could grow there" (Participants in the Refugee Settlement Study, 2025).

Due to the ability to hire external labor H_H , host households have a separable problem. This allows them to set total farm labor $L_H = F_H + H_H$ until the marginal revenue product of labor equals the wage rate w . This leads to a maximized profit function π_H^* and optimized labor demand L_H^* :

$$\pi_H = p_a L_H^\alpha \bar{A}_H^\beta - w L_H \implies \pi_H^* = p_a \beta \bar{A}_H \left(\frac{p_a \alpha}{w} \right)^{\frac{\alpha}{\beta}}; \quad L_H^* = \bar{A}_H \left(\frac{p_a \alpha}{w} \right)^{\frac{1}{\beta}} \quad (3)$$

Due to refugee household's inability to hire farm labor and fixed land size, their problem is non-separable. Their farm labor F_R is thus determined by the profit-maximizing condition, but their total time endowment is allocated among farm labor, off-farm market labor (S_R), and leisure. This leads to optimal production Q_R^* , labor allocation and consumption decisions to be simultaneously solved (see Appendix A). New arrival households N do not contribute to production in the current period; $Q_a^N = 0$.

2.1.2 Household Utility and Income

Households maximize a standard Cobb-Douglas utility function, which depends on consumption of the staple food crop, X_a , a composite non-food good, X_m and leisure X_l . The parameters γ, δ, θ represent

respective consumption preferences for each good and sum to one.:

$$U^i = \gamma \ln X_a + \delta \ln X_m + \theta \ln X_l \quad (4)$$

Household income m^i , comprises the total value of the time endowment (wT), farm profits (π^{*i}), and aid. Let Z_a^i and Z_m^i be the quantity of direct food relief aid and cash aid (in the form of vouchers) received by household i . The utility function $u^i(\cdot)$ is assumed to be strictly quasi-concave, ensuring a unique interior solution. The general budget constraint for all households is:

$$wT + \pi^i(p_a, w, \bar{A}^i) + Z_m^i + p_a Z_a^i = p_a X_a^i + X_m^i + wX_l^i \quad (5)$$

Households maximize utility (eq. 4) subject to their budget constraint (eq. 5), utilizing the total value of their time endowment wT , optimal farm earnings π^i and aid.

All aid, including Z_a^i , is tradable. The optimal consumption and labor supply decisions for each household type, as derived in Appendix A, are functions of the food price (p_a), the local wage (w), and their respective endowments ($\bar{A}_i$) and aid levels (Z_m^i, Z_a^i).

In-kind aid is allocated highest to new arrivals, followed by established refugees; $Z_a^N > Z_a^R > Z_a^H \geq 0$. For cash aid Z_m^i , established refugees receive the highest levels of cash transfers, followed by new arrivals and host households; $Z_m^R > Z_m^N > Z_m^H \geq 0$ (see Tables 1; B.3). This is consistent with local aid organizations promoting the transition of refugees from aid reliance to self-sufficient production (Mastrorillo et al., 2024).

2.1.3 Market Clearing

The model assumes that local markets for the staple food crop and labor clear. The equilibrium prices (p_a, w) are determined by equations 6 and 7:

$$\sum_{i \in \{H, R, N\}} N_i X_a^{*i}(p_a, w) = \sum_{i \in \{H, R\}} N_i Q_a^{*i}(p_a, w) + \sum_{i \in \{H, R, N\}} N_i Z_a^i \quad (6)$$

The left-hand side represents the aggregate market demand for the staple food crop by host, established and newly arrived refugee households. The term on the right-hand side represents the aggregate supply of the staple food crop by producing households (H and R) and the supply of direct food aid.

$$N_H H_H^* = N_R S_R^* + N_N S_N^* \quad (7)$$

The left-hand side of equation 7 represents the aggregate demand for hired labor inputs in the production model. The terms on the right-hand side represent the supply of off-farm provided labor for established refugee R and new arrival N households.

2.2 Price Effects and Predictions

The influx of refugees is modeled as an exogenous shock to the local economy through an increase in N_N . These new arrivals disturb the market equilibrium for both food and labor, $(\frac{dp_a}{dN_N}, \frac{dw}{dN_N})$, with the full comparative statics derived in the Appendix (Equations A.24 and A.25).

New arrivals likely increase local food prices, $(\frac{dp_a}{dN_N} > 0)$, through two mechanisms. The first is a direct increase in food demand F_{N_N} from new arrivals. Since new arrivals are non-producers, direct food aid Z_a^N acts as an intermediary, partially offsetting some of the new aggregate demand. This increase in F_{N_N} puts upward pressure on p_a . The second effect on food price comes through the wage induced production effects. New arrivals increase the labor supply, which puts downward pressure on wages w . This in turn incentivizes increased production on host and refugee plots, in turn, moderating food price effects. Overall, it is likely that the induced demand from new arrivals will outweigh the wage effects on production, leading to food price inflation from arrivals at the settlement level.

Wages are also likely to respond to new arrivals, which create a net depressing effect, $(\frac{dw}{dN_N} < 0)$. This is driven by two counteracting forces. Food price inflation, as described above, is likely to increase the marginal product of labor, increasing the demand for labor and exerting upward pressure on wages w . However, this effect is likely to be outweighed by the increase in labor supply brought on by sudden influxes of new arrivals. The wage-depressing effect may be attenuated if new arrivals receive higher amounts of cash aid (Z_m^N) , incentivizing increases in their demand for leisure (X_l^{*N}) and reducing their off farm labor supply. However, this may also encourage food-price inflation due to boosting demand for X_a^{*N} .

2.3 Food Security and Production Implications

The overall impact of refugee arrivals on local food security is indeterminate. The combination of increased demand, potential wage effects, and aid interventions makes the net effect on local food prices uncertain. If the upward pressure on prices from increased demand outweighs the downward pressures from increased supply and wage effects, then local food security could be negatively impacted. However, this demand increase could benefit households that are net producers. In contrast, if supply-side effects dominate, local food prices could fall, potentially improving food security.

Localized food price inflation would benefit net food sellers (typically some host farmers) but adversely affect net food buyers, which likely include many refugee households and poorer host households. The data in Table 1 summarizes basic characteristics of the host and refugee population in the sample (source further described below; Sec. 3) and suggests that refugee households, particularly new arrivals, produce less, earn less from agriculture, and are more likely to be net food consumers.

Conversely, if factors like increased local production due to lower wages or substantial in-kind food aid dominate, p_a might fall or rise less significantly. A decrease in p_a would generally improve food security for net consumers but reduce income for net sellers. This could have larger welfare implications for net consumers, which are more likely to be refugee households.

Food relief aid Z_a^i directly impacts the amount of food that a household needs to purchase from the market. It enters additively into the optimal food consumption (eq. (A.7), (A.15), (A.21)). The in-kind transfer relaxes the food budget constraint, while being a direct buffer against price inflation. For a given income m^i and price p_a , an increase in Z_a^i frees up resources that would have been spent on food, potentially increasing consumption of X_m^i or X_l^i , or allowing for higher overall food consumption X_a^i than would be possible without food aid.

Voucher or cash backed transfers (Z_m^i), enter the budget as a direct income transfer. This improves total consumption capacity for both food and non-food goods (eq.(A.8), (A.16), (A.22)). While these transfers may be beneficial, they are directly susceptible to localized inflation from new arrivals ($\frac{dp_a}{dN_N} > 0$). As p_a rises, the marginal benefit of Z_m^i to food consumption declines. Price inflation erodes the purchasing power of cash aid, making Z_m^i a less effective buffer against food insecurity than in-kind food transfers (Z_a^i) in this context.

Both aid modalities may also modify labor allocation and production decisions. The presence of food aid increases the non-labor component of income, reducing the need to allocate household labor towards on or off-farm production (eq. (A.10), (A.18), (A.23)) and incentivizing increases in leisure consumption. The effect on overall production is largely dependent on household land allocation and price effects.

3 Data

The main dataset is the Resilience Index Measurement and Analysis (RIMA) surveys conducted in Uganda between 2017 and 2021. The surveys are the result of a collaborative effort between the Ugandan Office of the Prime Minister and the Food and Agriculture Organization of the United Nations (FAO). They are designed to evaluate food security and resilience among refugee households within twelve settlements and among host households neighboring those settlements (FAO, 2019); see Figure 1. The data also includes various measures of agricultural production, infrastructure, demographic, and climatic exposure at the household level.

The RIMA survey employed a multi-stage stratified cluster sampling methodology. Stratification by household type (refugee/host) and region was designed to ensure representativeness. Clusters (within refugee settlements and nearby host villages) were randomly selected, followed by random selection of households within clusters for face-to-face interviews (FAO, 2019). Data were collected over four waves, creating a panel in which roughly 50% of households participated in all waves and 90% in at least two. The total sample comprises approximately 20,000 household-wave observations, with the distribution detailed in Appendix B.1. I restrict the sample to households present across all waves, resulting in a final subset of approximately 9,000 observations (see Table 1) in a balanced panel.

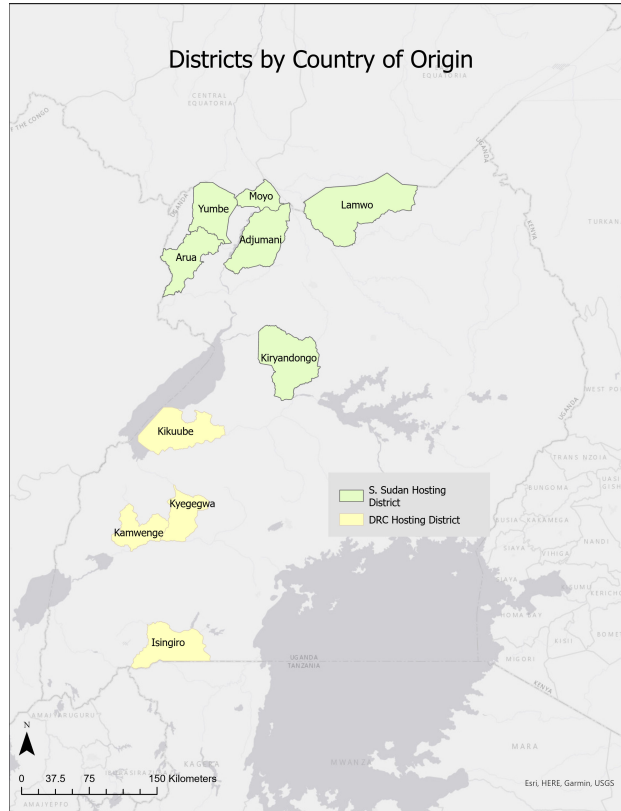


Figure 1: Location of Districts with Settlements and their Source of Refugees

The central focus of this study is the Coping Strategies Index (CSI), which measures the level of food insecurity based on a weighted index of various coping strategies households may have used within a given week (FAO, 2019). Elements include household experiences and behaviors, such as food rationing, the reduction of meal frequency, and the selling of assets.

Table 1 presents a summary of household characteristics and food security levels, contrasting refugees and host households for the entire panel. The table reveals significant disparities between the two groups. Refugee households generally exhibit higher levels of food insecurity, as evidenced by higher CSI scores, and possess

fewer assets and total heads of livestock when compared to host households. Furthermore, refugees typically cultivate smaller land areas and exhibit lower agricultural output. Notably, refugee households receive higher levels of food voucher and direct food aid. The characteristics of the full sample as well as refugee households by number of observations across panel waves can be seen in Tables B.2 and B.3. Observable characteristics are broadly similar across both the balanced and full panels, suggesting that attrition is not strongly selective on baseline observables. It is important to note that attrition from the settlement is paired with a high fixed cost: forfeiture of access to allocated land, food rations, voucher transfers and social services. Therefore, attrition in this analysis is more likely to be driven by survey logistics than by household flight.

Table 1: Summary Statistics: Refugee and Host Households

	Refugees	Hosts	Normalized Difference
<i>Panel A: Outcomes</i>			
Coping Strategies Index	25.95	16.51	0.33
Any Harvest	0.86	0.96	-0.25
Harvest (KG)	201.56	986.78	-0.52
Any Agricultural Earnings	0.34	0.64	-0.44
Agricultural Earnings (USD)	12.88	104.67	-0.40
Total Crops Sold (KG)	52.57	310.45	-0.39
Any Livestock Ownership	0.43	0.70	-0.40
Livestock Holdings (Heads)	2.01	3.98	-0.31
<i>Panel B: Household Characteristics</i>			
Years of Schooling	8.21	8.05	0.02
Household Size	6.34	6.57	-0.05
Total Land Holdings (Acres)	0.56	4.03	-0.64
Normalized Asset Index	-0.34	0.51	-0.66
Voucher Aid (USD)	169.52	7.59	0.66
Relief Aid (USD)	127.04	1.51	0.59
Months in Settlement	201.15	.	.
Observations	4,267	4,685	.

Notes: The table reports sample means for refugee and host households. The normalized difference is defined as the difference in means divided by the square root of the average of group variances.

Refugee arrival data is sourced from the United Nations High Commissioner for Refugees (UNHCR) via the Uganda Comprehensive Refugee Response Portal. This dataset provides monthly records of the total refugee population in each settlement from 2016 to 2022, although data availability for some districts during the early years (2016-2017) is limited. Figure 2 illustrates the temporal variation in refugee arrivals across different districts. In the arrival data, all district-level observations show arrivals increasing throughout the available survey period.

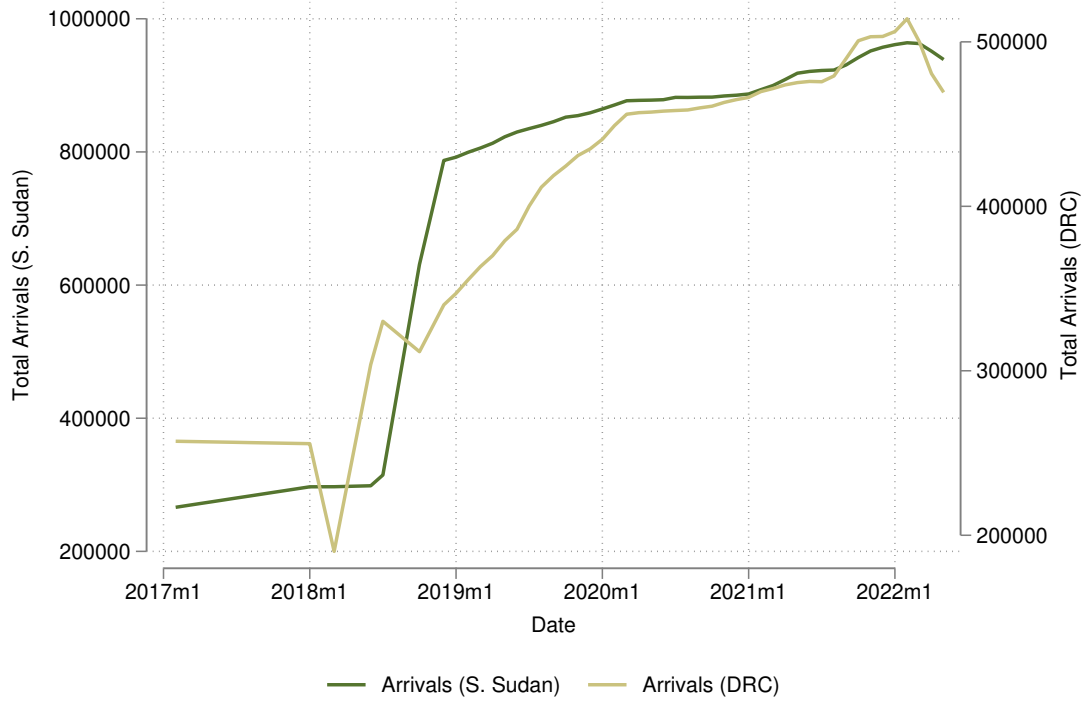


Figure 2: Total Arrivals by Country of Origin

Prices come from the World Food Programme price database (“Uganda - Food Prices”, 2025) which tracks monthly prices for various commodities at the refugee settlement and disparate market level. For estimation I focus on five staple commodities with the most complete series, which are a combination of aid and non-aid goods. Two of the aid commodities, beans and maize, are both produced within settlements, while oil, the third commodity has minimal production within settlements (WFP, 2023). The non-aid goods millet and sorghum are both produced at the settlement level, and are preferred carbohydrate sources for South Sudanese refugees while not being widely consumed by refugees from the DRC (Willis & Buck, 2007). The dates of the price panel extend from January of 2021 to January of 2023, only aligning with the household panel for the year of 2021. Average prices for each settlement characteristic are reported in Table B.4 with the changes in standardized prices in relation to average settlement arrivals visualized in Figure 3. For an overview of prices at the settlement level see Table B.5; and for a visualization of individual commodity prices see Figure B.1. Prices between settlements and commodities are largely similar.

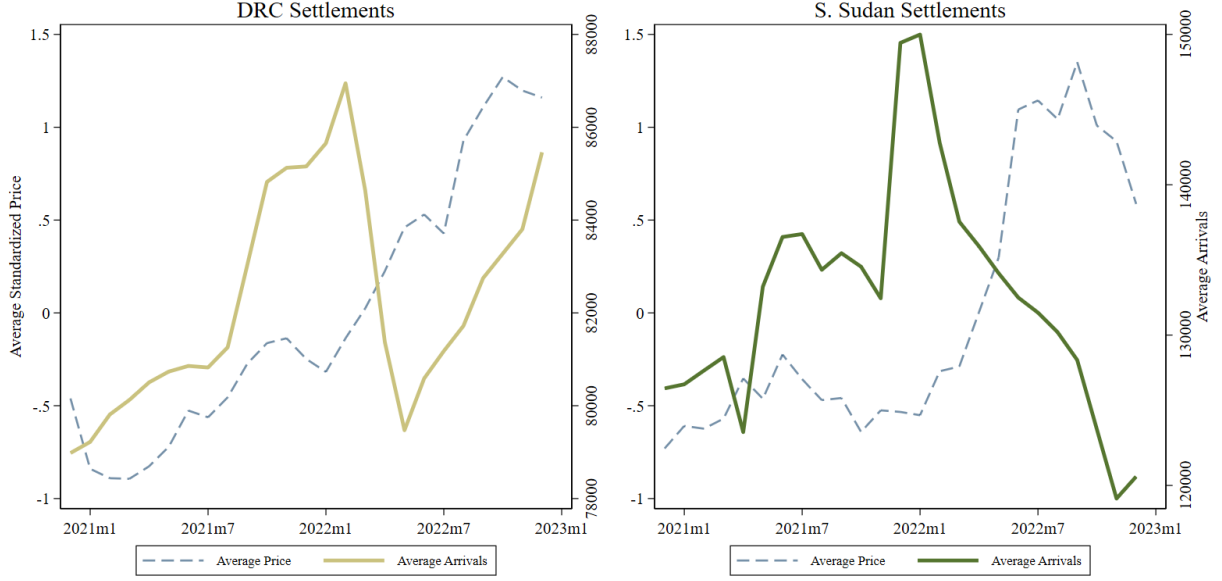


Figure 3: Average Commodity Prices and National Arrivals by Settlement Type

To incorporate the influence of conflict as a driver of displacement, armed conflict data for the Democratic Republic of the Congo (DRC) and South Sudan (the primary sources of refugees in Uganda during the study period) is obtained from the Armed Conflict Location & Event Data Project (ACLED) for the period of 2016-2022. This dataset captures the frequency, location, and nature of violent events, drawing upon a combination of citizen reports and news sources (Raleigh et al., 2023). Figure 4 shows conflict points and the host districts for the respective countries. I filter observations to include only those within 500km of the Ugandan border. This distance is based on the recorded journeys of displaced persons in South Sudan. (Talleraas et al., 2025). The spatial distribution of conflicts indicates that settlements receiving refugees from the DRC are generally closer to conflict zones, whereas displacement from South Sudan is associated with a more dispersed conflict pattern. Additionally, there are notable spikes in violence within the study period driving the increasing number of arrivals over time.

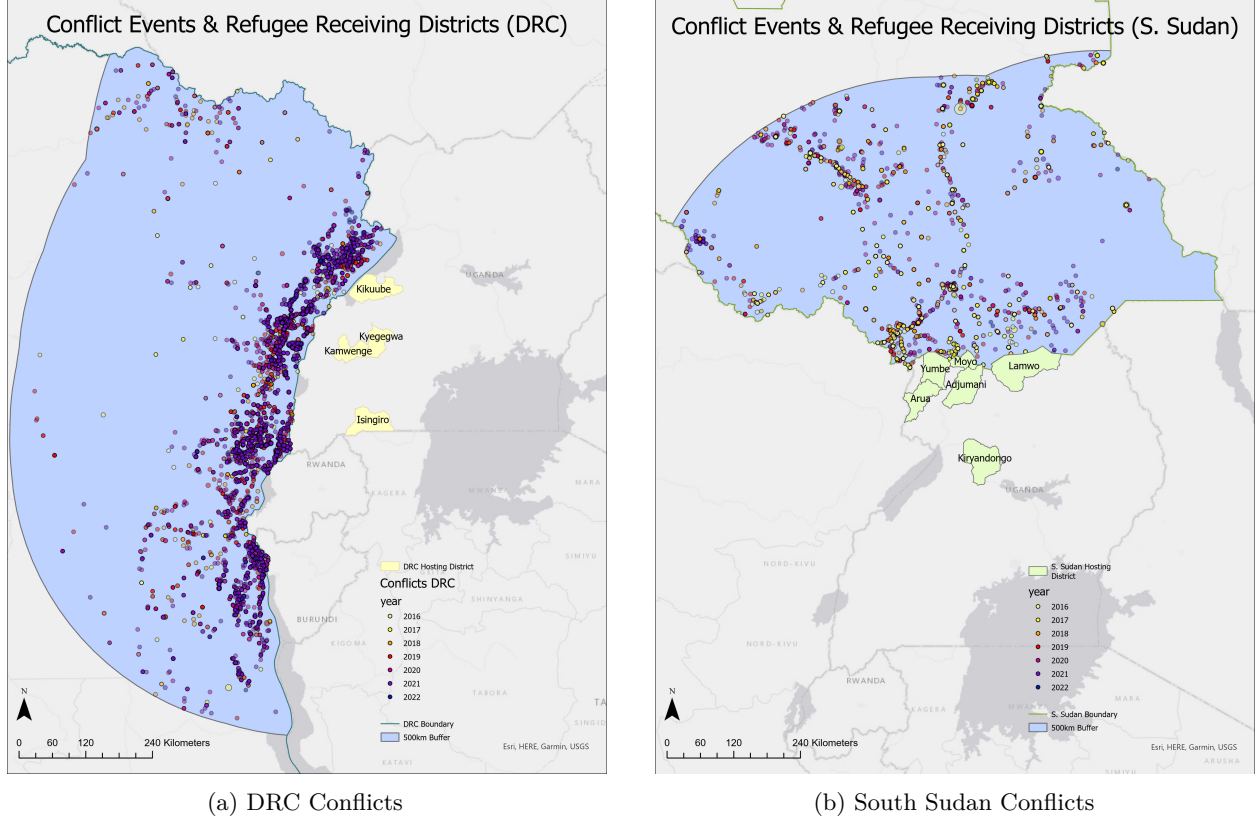


Figure 4: Conflict Events in Country of Origin and Respective Host Districts

The World Bank’s Climate Change Knowledge Portal provides monthly data on various climatic variables, including temperature, precipitation, and drought indices, at a 0.5-degree spatial resolution (World Bank, 2024). All datasets for household analysis are combined at the district-level to create a cohesive panel used to estimate responses from the increase in arrivals seen in the time period.

4 Household Estimation

This section outlines the empirical strategy employed to estimate the causal impact of refugee arrivals on food security and production outcomes in designated settlement communities. Estimation of the effects of refugee arrivals on food security and production outcomes leverages variation in refugee arrivals across settlements and over time.

The equation of interest is:

$$Y_{ijt} = \delta_1 \cdot \Delta Arrivals_{jt-l} + W_{jt}\Gamma + X_{ijt}\Lambda + \mu_t + \theta_i + \epsilon_{ijt} \quad (8)$$

Y_{ijt} is the outcome, either the Coping Strategies Index (CSI) score or various agricultural production decisions for household i , in district j , during year t . $\Delta Arrivals_{jt-l}$ represents the change in refugee population for the district from the lag l to the current time period t . Various lags —3, 6 and 12 months—account for the delayed effects of refugee arrivals due to processing times, production lags, and the gradual reduction of per-capita aid over time for these new arrivals. W_{jt} is a vector of climate controls for district j , intended to capture contemporary and seasonal agricultural conditions. Due to Uganda’s two annual growing seasons and the survey’s staggered interview dates, these variables are constructed from a 6-month rolling window of climate data ending in the survey month t . The vector includes the mean, maximum, and minimum temperatures and cumulative precipitation over the six-month period, along with their respective quadratics to account for nonlinearities. This rolling window approach helps approximate the contemporaneous climate, along with the growing conditions preceding the outcomes measured. X_{ijt} is a vector of time-varying household survey controls, specifically the household size, tenure in the settlement and the value of relief and seed aid received over the past year.¹ The use of two-way fixed effects are included to capture time-invariant household characteristics θ_i and year fixed effects μ_t . ϵ_{ijt} is the error term. To account for the nature of treatment —the arrival of refugees at the settlement level —standard errors are clustered at the district level (Abadie et al., 2017). I define the district as the effective catchment area, as it encompasses both host and established refugee households residing within or in close proximity to the settlements. Due to clustering, there exists a small N problem (11 districts). To address this I apply the wild cluster bootstrap (Davidson & Flachaire, 2008).

4.1 Identification Strategy

Estimating the causal effect from equation 8 may result in a downward bias for δ_1 due to endogeneity. Refugees are not randomly assigned to settlements. Instead assignments are based on perceived land availability, aid, settlement crowding and overall absorption capacity (Zhu et al., 2024). The nature of this arrival system may lead to refugees being assigned, or self-selecting to camps that have better food security at the time of their arrival.

To address this issue, I employ an instrumental variable (IV) strategy using a modified shift-share instrument. The instrument aims to isolate exogenous variation in refugee arrivals driven by push factors in the countries of origin, channeled through settlement characteristics.

¹In the Ugandan settlement context, household-level aid allocations are determined by national funding constraints and a predetermined phase-out schedule based on a household’s tenure, rather than acting as a localized response to contemporaneous arrival shocks (Brown & Torre, 2024). National budgets for refugee management also remained largely stable over the study period (UNHCR, 2025).

I construct the instrument, $\Delta Arrivals_{jt}^{IV}$, to predict the change in the refugee population, $\Delta Arrivals_{jt}$. The instrument interacts measures of conflict intensity and total outflow in the country of origin (the "shift") with characteristics of the receiving settlement areas and households (the "share").

Let o denote the primary country of origin (South Sudan or DRC) for refugees arriving in district j . The first stage is constructed as:

$$\begin{aligned} \Delta Arrivals_{jt}^{IV} = & \pi_0 + \pi_1 \underbrace{\left(\frac{Area_j}{Distance_j} \right)}_{\text{Share component}} \times \underbrace{(Deaths_{oj(t-6)} \cdot TotalArrivals_t)}_{\text{Shift component}} \\ & + W_{jt}\Pi + X_{ijt}\Phi + \mu_t + \theta_i + u_{ijt} \end{aligned} \quad (9)$$

The "shift" portion of the instrument captures push factors. $Deaths_{oj(t-6)}$ is the total number of conflict-related fatalities reported by ACLED within a 500km buffer zone inside country of origin o , occurring 6 months prior to period t . The lag and buffer zone are based on estimates of average journey times and distances for displaced persons in the South Sudan (Talleraas et al., 2025). $TotalArrivals_t$ is the total number of refugee arrivals across all receiving districts in Uganda in time t . This term scales the conflict intensity with the overall inflow from the origin country in that period.

The "share" component captures settlement-specific pull factors and capacity. $Area_j$ is the size (in km²) of the UNHCR-designated settlement planning boundary in district j . $Distance_j$ is the Euclidean distance (in km) from the settlement in district j to the nearest point on the border of country of origin o . Exposure to refugee shocks is therefore modeled exclusively by the settlement characteristics and total arrivals.

The first stage estimation also includes the same set of controls and fixed effects in the main estimation, $W_{j,t-l}$, X_{ijt} , μ_t , and θ_i . u_{ijt} is the first-stage error term. The interaction aims to capture the idea that exogenous shocks (conflict driving total outflows) lead to larger changes in arrivals in districts that are larger and closer to the border when controlling for household characteristics.

4.2 Relevance & Validity

The relevance of the instrument is demonstrated visually in Figure 5. The total number of conflict deaths from six months prior is strongly correlated with the aggregate number of arrivals. The data also demonstrate that districts closer to the border and those containing larger settlement boundaries have larger displaced populations overall. I formally test for instrument relevance by reporting the Kleibergen-Paap rk Wald F-statistic from the first-stage regression in all main results.

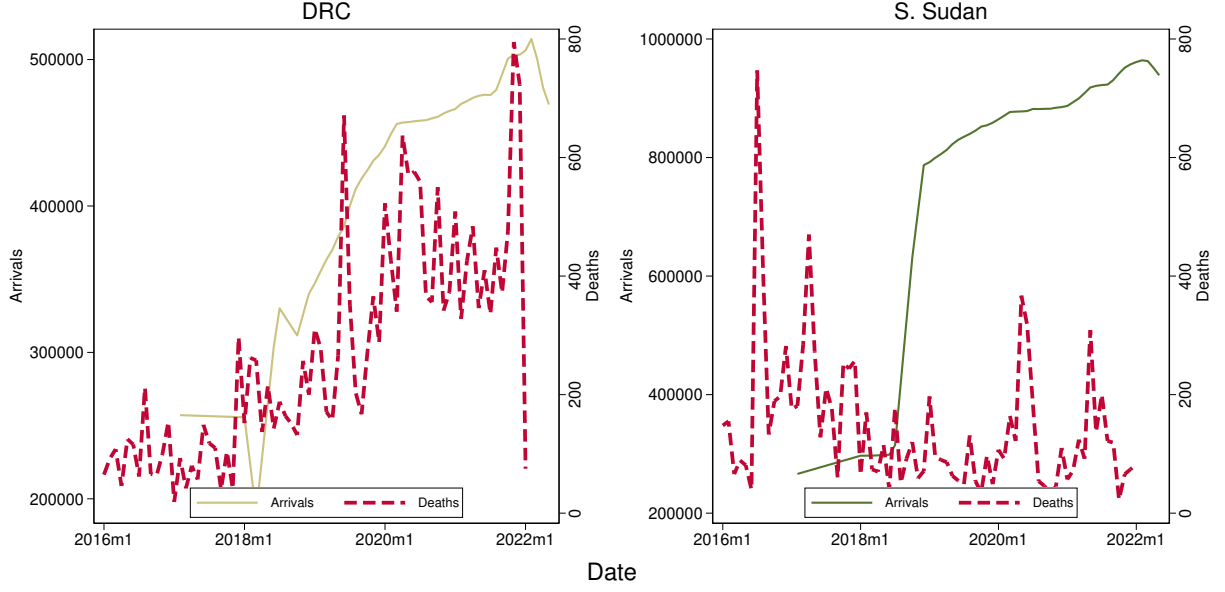


Figure 5: Refugee Arrivals by District Type and Deaths in Country of Origin

For the instrument to be valid, it must affect the outcome of interest (food security, production decisions) only through its effect on $\Delta Arrivals_{jt-l}$, and not through any unobserved channel conditional on controls and included fixed effects. This assumption is untestable. However, I argue below the push factors (conflict intensity and aggregate arrivals) and pull factors (area and distance) are exogenous to the household production and consumption decisions in Uganda.

The first concern is that the share component —settlement distance to the country-of-origin border and baseline settlement area—may be correlated with pre-existing local conditions such as market integration, agro-ecological characteristics, infrastructure, or remoteness. If these local factors cause districts to exhibit differential responses over time, independent of refugee arrivals, the exclusion restriction may be violated. The specific mechanism required for such a violation is that border proximity or settlement size affects within-household changes in food security through channels other than the refugee arrivals they induce. I argue this channel is implausible. I formally test for the presence of differential trends in Appendix C, finding none.

A secondary threat to validity is the potential for direct, cross-border spillovers from the conflict itself (the shift component). Increased border-area conflict ($Deaths_{oj(t-6)}$) could directly disrupt cross-border food trade, impacting local prices and food security independently of refugee inflows. However, given the remote nature of the settlements and the relatively small scale of cross-border agricultural trade in these areas, it is highly unlikely that regional agricultural markets are significantly impacted by these specific conflict events.

National trade statistics corroborate this assumption. Uganda’s food supply is not meaningfully dependent on trade with South Sudan or the DRC. The Observatory of Economic Complexity (OEC) (2023) reports that between 2017 and 2023, imports from South Sudan accounted for only 0.21% of Uganda’s total imports, with crops comprising just 1.6% of that minor share. Imports from the DRC were similarly negligible, representing 0.15% of total imports and including almost no agricultural products. While Uganda’s crop *exports*—13% of the 6% of total exports to South Sudan, and 3.68% of the 4.9% to the DRC—are somewhat larger, the small levels of overall trade and comparatively minimal import flows provide evidence that conflict-led trade disruptions would have a negligible impact on settlement food supply and prices.

There is also the fact that the persistent level of violence in border regions renders robust regional markets unlikely. Settlement interviews confirm widespread perceptions that cross-border trade is unsafe and not feasible. Refugees repeatedly emphasized fears about returning to their home countries, a sentiment justified by reports of returnees being killed. One refugee explicitly stated, "It is not possible to trade. People are not authorized to move across the border... now it is illegal and very risky". A settlement commandant similarly described the situation in South Sudan as "a barrel that is still smoking," explaining the low likelihood of refugees returning, let alone engaging in regular trade (Participants in the Refugee Settlement Study, 2025).

Finally, there is a possibility that spatial conflict spillovers could impact agricultural output directly. However, evidence suggests that rural settlements are significantly less affected by direct conflict spillovers compared to their urban counterparts (Fein & Van Den Hoek, 2022).

Given the included fixed effects and time-varying controls, and absent direct conflict or trade spillovers, the instrument isolates variation in refugee arrivals driven by the interaction between exogenous push factors and settlement characteristics. To further validate this, I conduct two formal falsification exercises—a temporal placebo test using forward observations of arrivals, and a leave-one-district-out estimation—to assess the plausibility of differential trends and to ensure outsized impacts are not driven by a single influential district. Both exercises, reported in Appendix C, strongly support a causal interpretation of the main findings.

5 Household Results

This section presents the estimation results for household-level outcomes, first focusing on food security and agricultural production (Sections 5.1 & 5.2). I discuss findings for the preferred IV specification for both host and established refugee households, with first-stage results, and baseline OLS/alternative fixed-effect specifications reported in the Appendix (Tables B.6-B.23). I then explore the heterogeneous impacts of arrivals based on household market participation in Section 5.3 and summarize the robustness and validity of outcomes in Section 5.4.

Table 2
Host Households: Impact of Refugee Arrivals on Food Security and Production

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: 3 Month Lag</i>								
Arrivals (3 Month Lag)	0.113 (0.072)	0.019** (0.012)	0.056 (0.114)	0.012 (0.032)	0.092 (0.146)	0.058 (0.200)	-0.030 (0.037)	-0.148** (0.087)
Mean Dep. Var.	0.20	0.96	5.97	0.63	2.62	3.30	0.70	1.30
Observations	3386	3357	3226	3357	3211	3228	3357	3293
R-squared	0.027	0.020	0.066	0.017	0.025	0.023	0.145	0.100
Weak ID (Wald F)	10.01	9.43	9.23	9.43	9.57	9.53	9.43	9.14
UnderID P-Value	0.034	0.037	0.038	0.037	0.038	0.037	0.037	0.039
Wild Boot. p-val	0.285	0.045	0.648	0.793	0.639	0.819	0.359	0.049
<i>Panel B: 6 Month Lag</i>								
Arrivals (6 Month Lag)	-0.113** (0.056)	0.023 (0.018)	0.168 (0.145)	0.064** (0.038)	0.230 (0.182)	0.315 (0.227)	0.057*** (0.024)	0.158** (0.064)
Mean Dep. Var.	0.20	0.96	5.97	0.63	2.62	3.30	0.70	1.30
Observations	3386	3357	3226	3357	3211	3228	3357	3293
R-squared	-0.048	-0.056	-0.014	-0.053	-0.021	-0.041	0.120	0.036
Weak ID (Wald F)	5.69	5.80	5.61	5.80	5.64	5.72	5.80	6.00
UnderID P-Value	0.016	0.015	0.016	0.015	0.016	0.016	0.015	0.015
Wild Boot. p-val	0.043	0.228	0.228	0.048	0.222	0.195	0.000	0.026
<i>Panel C: 12 Month Lag</i>								
Arrivals (12 Month Lag)	-0.053** (0.024)	0.010 (0.008)	0.074 (0.057)	0.029* (0.015)	0.103 (0.076)	0.139 (0.090)	0.027*** (0.010)	0.074** (0.028)
Mean Dep. Var.	0.20	0.96	5.97	0.63	2.62	3.30	0.70	1.30
Observations	3386	3357	3226	3357	3211	3228	3357	3293
R-squared	-0.054	-0.041	-0.001	-0.053	-0.023	-0.038	0.107	0.028
Weak ID (Wald F)	10.68	10.68	10.40	10.68	10.32	10.93	10.68	11.07
UnderID P-Value	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
Wild Boot. p-val	0.029	0.243	0.234	0.081	0.235	0.218	0.000	0.033
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables correspond to household food security and various agricultural production outcomes, as indicated in the column headers. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months. All models are estimated using our preferred specification of year and household fixed effects and a Shift-Share IV. Standard errors, reported in parentheses, are clustered at the district level. To account for the small number of clusters, p-values from a wild cluster bootstrap (using Webb weights) are reported. When the Weak Identification Wald F-statistic falls below the critical threshold of 16.38, the reported bootstrap p-values incorporate an Anderson-Rubin (AR) test, which is robust to weak identification. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3
Refugee Households: Impact of Arrivals on Food Security and Production

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: 3 Month Lag</i>								
Change in Arrivals	0.184 (0.161)	0.006 (0.035)	0.271 (0.294)	-0.089 (0.055)	-0.076 (0.155)	-0.081 (0.228)	0.085 (0.083)	0.151 (0.143)
Mean Dep. Var.	-0.32	0.85	3.81	0.28	0.71	1.02	0.48	0.78
Observations	2965	3020	2793	3020	2799	2793	3020	2986
R-squared	0.051	0.146	0.043	0.040	0.004	0.006	0.463	0.432
Weak ID (Wald F)	19.13	18.47	20.57	18.47	20.33	20.57	18.47	18.76
UnderID P-Value	0.052	0.056	0.041	0.056	0.041	0.041	0.056	0.056
Wild Boot. p-val	0.810	0.946	0.504	0.426	0.708	0.773	0.480	0.463
<i>Panel B: 6 Month Lag</i>								
Change in Arrivals	-0.095** (0.052)	0.023 (0.025)	0.139 (0.121)	0.023* (0.015)	0.062 (0.040)	0.062 (0.065)	0.000 (0.022)	-0.011 (0.044)
Mean Dep. Var.	-0.32	0.85	3.81	0.28	0.71	1.02	0.48	0.78
Observations	2965	3020	2793	3020	2799	2793	3020	2986
R-squared	0.030	0.106	0.009	0.036	0.009	0.012	0.483	0.456
Weak ID (Wald F)	6.80	7.51	7.86	7.51	7.98	7.86	7.51	7.64
UnderID P-Value	0.018	0.020	0.019	0.020	0.019	0.019	0.020	0.020
Wild Boot. p-val	0.024	0.580	0.461	0.073	0.229	0.456	0.993	0.884
<i>Panel C: 12 Month Lag</i>								
Change in Arrivals	-0.050** (0.025)	0.010 (0.011)	0.060 (0.055)	0.012* (0.007)	0.031 (0.019)	0.032 (0.029)	-0.002 (0.011)	-0.008 (0.022)
Mean Dep. Var.	-0.32	0.85	3.81	0.28	0.71	1.02	0.48	0.78
Observations	2965	3020	2793	3020	2799	2793	3020	2986
R-squared	0.004	0.118	0.018	0.044	0.009	0.013	0.484	0.456
Weak ID (Wald F)	11.97	16.06	17.02	16.06	17.24	17.03	16.06	16.49
UnderID P-Value	0.013	0.018	0.020	0.018	0.020	0.020	0.018	0.018
Wild Boot. p-val	0.012	0.581	0.547	0.087	0.106	0.342	0.909	0.828
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables correspond to household food security and various agricultural production outcomes, as indicated in the column headers. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months. All models are estimated using our preferred specification of year and household fixed effects and a Shift-Share IV. Standard errors, reported in parentheses, are clustered at the district level. To account for the small number of clusters, p-values from a wild cluster bootstrap (using Webb weights) are reported. When the Weak Identification Wald F-statistic falls below the critical threshold of 16.38, the reported bootstrap p-values incorporate an Anderson-Rubin (AR) test, which is robust to weak identification. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.1 Household Food Security Outcomes

Column (1) of Tables 2 and 3 present the estimated effects of refugee arrivals on the food security of host and established refugee households, respectively. The tables report the shift-share instrumental variables (IV) estimates from the preferred specification outlined in Section 4.1. The outcome variable is the standardized Coping Strategies Index (CSI). Unadjusted, a higher CSI score indicates a greater reliance on negative coping strategies (worse food security). For ease of interpretation, I reverse the standardized index such that lower values correspond to worse food security. Therefore, a negative coefficient in the estimations indicates a deterioration in household food security, measured in standard deviations. The variable of interest is the change in the refugee population, $\Delta Arrivals_{j(t-l)}$, measured in thousands and evaluated at lags (l) of 3, 6, and 12 months (Panels A, B, and C). First-stage results are reported in Tables B.6 and B.7, while alternative OLS, fixed effects, and robustness specifications are reported in Appendix Tables B.8 and B.9.

The preferred specifications include year fixed effects (μ_t), household fixed effects (θ_i), and controls for household size, tenure, lagged seasonal precipitation and temperature (6-month lags) with their respective quadratic terms, and a lagged measure of household relief aid (USD). I report the Kleibergen-Paap rk LM statistic for under-identification and the Kleibergen-Paap Wald rk F-statistic for weak identification. To account for the small number of clusters ($G < 11$), I report wild cluster bootstrap p-values using Webb weights. When the first-stage F-statistic falls below the Stock–Yogo 10% critical value of 16.38, statistical significance is based on the weak-instrument-robust Anderson-Rubin (AR) test (Davidson & Mackinnon, 2010; Roodman et al., 2019).

The results suggest that refugee arrivals cause a statistically significant deterioration in food security for both host and established refugee households over the medium and long term. While short-run (3-month) effects are statistically insignificant, the adverse effects at both the 6-month and 12-month lags are significant at the 5% level and robust to the wild cluster bootstrap p-values. As noted at the bottom of each panel, while the instrument is generally robust, the Wald F-statistic for the clustered specifications at the 6-month lag falls below 10 for both groups. However, because inference relies on the weak-instrument-robust AR wild cluster bootstrap p-values, these lower F-statistics do not substantively compromise the interpretation of the estimates (Staiger & Stock, 1997).

The magnitude of these effects is economically meaningful. At the 12-month lag, a 1,000-person increase in refugee arrivals reduces the standardized food security score by 0.0597 SD for hosts and 0.0477 SD for established refugees. Contextualizing this with the original CSI scores (Table 1), leads to an absolute *increase* in the coping strategies index of 0.99 points for hosts and 1.23 points for refugees per 1,000 arrivals. Scaling by the average annualized district-level arrivals (9,182 persons), hosts experience an average 9.1-point increase

in their CSI, while refugees experience an 11.37-point increase. These shifts are severe: they move the average host household (baseline raw CSI = 16.51) and the average refugee household (baseline = 25.95) into the bottom quartile of their respective baseline food security distributions.

The medium-term (6-month lag) estimates imply a similar worsening of food security, with an increase in CSI of 2.1 points for hosts and 2.33 points for refugees, on average. Scaled to the average increase in arrivals in the 6 month period (3,197), sees host households experience roughly a 6.7 point increase in food insecure behavior, while refugees undergo a 7.46 point increase. These CSI increases push the average host household to the 26th percentile and the average refugee household to the 28th percentile of baseline food security.

Appendix Tables B.8 and B.9 report the full set of alternative specifications. Column (1) presents baseline OLS estimates, which generally indicate attenuated or positive associations between arrivals and food security. Columns (2) and (3) add year and district fixed effects, while Columns (4) and (5) incorporate household fixed effects. Columns (6) and (7) report IV estimates using the shift-share instrument, with Column (7) additionally reporting wild cluster bootstrap p-values. It is important to note that these naive results in comparison to the preferred shift-share estimations—which show a negative effect on food security—suggests that refugee arrivals are sorted into settlements with greater initial food security and absorptive capacity. Across both host and refugee samples, the main specification finds adverse medium and longer-run effects of refugee arrivals on household food security.

5.2 Agricultural Production Outcomes

I also investigate the effects of refugee arrivals on various household agricultural production outcomes, focusing specifically on market participation and livestock ownership. Tables 2 and 3 present these results for host and established refugee households, respectively. The tables report the extensive and intensive margins of crop harvesting (Columns 2 and 3), agricultural market earnings (Columns 4 and 5), the volume of crops sold (Column 6), and livestock holdings (Columns 7 and 8). The independent variable remains refugee arrivals (in thousands) over the preceding 3, 6, and 12 months (Panels A, B, and C). For all outcomes, model specifications and estimation strategies mirror those used for the food security analysis. First-stage results are again reported in Tables B.6 and B.7, while all alternative specifications are reported in Tables B.10 through B.21.

The results indicate that refugee inflows induce significant shifts toward market entry for both host and refugee households, particularly in the medium and longer term. For host households, the preferred IV specification reveals increased market participation at the 6- and 12-month lags. Specifically, at the 6-month mark, a 1,000-person increase in arrivals increases the likelihood of reporting any annual agricultural earnings

by 6.42 percentage points. A similar effect persists at 12 months, with a 2.93 percentage point increase ($p < 0.10$).

To interpret the magnitude, I scale these coefficients by the average district-level arrivals (3,197 and 9,182 over the 6- and 12-month windows, respectively). These estimates imply that refugee inflows increase the probability that an average host household engages in market sales by 20.5 and 26.6 percentage points. Given the average rate of any earnings (63 percent), this represents a substantial shift toward market participation for host households.

For established refugee households, the IV estimates similarly demonstrate evidence of increased market participation. A 1,000-person increase in refugee arrivals raises the likelihood of reporting any earnings by 2.25 and 1.22 percentage points at the 6- and 12-month lags, respectively ($p < 0.10$). Scaled to the average district-level arrivals, established refugee households on average increase their probability of reporting earnings by 7.19 and 11.2 percentage points. Established refugees, although exhibiting a much lower baseline rate of market participation (28 percent), still show large movement towards participation induced by refugee arrivals. The shock of inflows triggers similar responses between household types: a movement away from pure subsistence (supplemented by aid for refugee households) towards market engagement.

Beyond crop market extensification, there are significant adjustments in livestock holdings—a form of liquid wealth and capital investment for rural Ugandan households (Behnke & Nakirya, 2012)—across both the extensive and intensive margins (Table 2, Columns 7 and 8). For host households, the immediate short-run response (3-month lag) to arrivals is a significant reduction in the intensive margin of livestock (a 16.28 percent decrease, when scaled to average arrivals in the period). This suggests a liquidation of assets in response to the initial influx, potentially used for further capitalization. This is reflected in the medium to long term (6- and 12-month lags). Host households' accumulation of livestock is significant for both the likelihood of any livestock and the total number of heads. When scaled to average arrivals in the 6-month period (3,197), hosts experience a 17.67 percentage point increase in the likelihood of owning livestock and a 50.5 percent increase in their total heads of livestock. Scaling the 12-month lag findings (9,182 arrivals), finds that hosts increase their likelihood of ownership by 24.8 percentage points and their heads of livestock by 68 percent.

In contrast, established refugee households exhibit no statistically significant response in livestock holdings across any time horizon (Table 3, Columns 7 and 8). This divergence highlights bifurcated outcomes: while host households possess the initial capital to utilize livestock for beneficial market entry and wealth accumulation, refugee households likely face higher initial constraints, preventing them from adopting similar wealth-accumulating behaviors.

The improvements in the agricultural market participation and livestock ownership (for Hosts) combined with the null results found for total harvests, contrast with the negative impacts found on food security in Section 5.1. This suggests that the primary mechanism driving increased food insecurity—particularly at the 6- and 12-month marks—is localized price effects (see Sec. 7) without a corresponding expansion in non-market food sources such as aid. This pattern indicates that households are likely selling agricultural goods to improve liquidity without expanding production, a coping strategy in the face of localized inflation. Long run effects are likely to be driven by the household’s initial capital. I investigate this heterogeneity by household type and baseline production characteristics in Section 5.3.

5.3 Heterogeneous Outcomes by Market Participation

Table 4
Host Households: Heterogeneous Impacts by Baseline Market Participation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: Consumers (6 Month Lag)</i>								
Arrivals (6 Month Lag)	-0.164 (0.049)	0.024 (0.023)	0.146 (0.168)	0.049 (0.050)	0.244 (0.239)	0.296 (0.297)	0.066* (0.037)	0.181 (0.103)
Mean Dep. Var.	0.11	0.96	5.67	0.54	2.08	2.62	0.76	1.43
Observations	1137	1125	1083	1125	1087	1091	1125	1100
R-squared	-0.017	0.029	0.069	0.031	0.017	0.025	0.076	-0.023
Weak ID (Wald F)	9.10	9.45	9.81	9.45	9.87	9.79	9.45	9.75
UnderID P-Value	0.018	0.017	0.015	0.017	0.016	0.015	0.017	0.016
Wild Boot. p-val	0.174	0.499	0.562	0.383	0.424	0.308	0.060	0.168
<i>Panel B: Consumers (12 Month Lag)</i>								
Arrivals (12 Month Lag)	-0.082*** (0.022)	0.010 (0.010)	0.061 (0.075)	0.022 (0.024)	0.112 (0.114)	0.133 (0.137)	0.032 (0.017)	0.085 (0.048)
Mean Dep. Var.	0.11	0.96	5.67	0.54	2.08	2.62	0.76	1.43
Observations	1137	1125	1083	1125	1087	1091	1125	1100
R-squared	-0.023	0.038	0.066	0.024	0.000	0.009	0.062	-0.042
Weak ID (Wald F)	13.24	13.24	13.78	13.24	13.43	13.63	13.24	13.59
UnderID P-Value	0.017	0.017	0.016	0.017	0.017	0.016	0.017	0.016
Wild Boot. p-val	0.003	0.607	0.737	0.478	0.468	0.422	0.331	0.282
<i>Panel C: Producers (6 Month Lag)</i>								
Arrivals (6 Month Lag)	-0.081 (0.077)	0.028 (0.024)	0.240 (0.210)	0.079*** (0.047)	0.250 (0.213)	0.369** (0.267)	0.055*** (0.034)	0.166*** (0.088)
Mean Dep. Var.	0.25	0.97	6.12	0.68	2.90	3.65	0.68	1.24
Observations	2249	2232	2143	2232	2124	2137	2232	2193
R-squared	-0.035	-0.123	-0.107	-0.121	-0.044	-0.089	0.138	0.051
Weak ID (Wald F)	3.27	3.23	3.12	3.23	3.12	3.27	3.23	3.34
UnderID P-Value	0.019	0.018	0.021	0.018	0.019	0.019	0.018	0.016
Wild Boot. p-val	0.232	0.138	0.165	0.000	0.148	0.045	0.009	0.000
<i>Panel D: Producers (12 Month Lag)</i>								
Arrivals (12 Month Lag)	-0.037 (0.036)	0.013 (0.010)	0.111 (0.085)	0.037*** (0.020)	0.114 (0.090)	0.167* (0.109)	0.026** (0.016)	0.079*** (0.041)
Mean Dep. Var.	0.25	0.97	6.12	0.68	2.90	3.65	0.68	1.24
Observations	2249	2232	2143	2232	2124	2137	2232	2193
R-squared	-0.044	-0.114	-0.079	-0.120	-0.041	-0.076	0.124	0.043
Weak ID (Wald F)	4.41	4.32	4.38	4.32	4.31	4.69	4.32	4.53
UnderID P-Value	0.018	0.018	0.018	0.018	0.017	0.017	0.018	0.017
Wild Boot. p-val	0.258	0.136	0.158	0.000	0.190	0.058	0.011	0.003
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables correspond to household food security and various agricultural production outcomes. The sample is split into "Consumers" and "Producers" based on any market participation (agricultural earnings) in the first observation period. The independent variable is the change in refugee arrivals (in thousands) over the preceding 6 or 12 months. All models are estimated using a Shift-Share IV with year and household fixed effects. Standard errors in parentheses are clustered at the district level. P-values from a wild cluster bootstrap (using Webb weights) are reported to account for the small number of clusters. For Weak Identification Wald F-statistics below 10.00, the bootstrap p-values incorporate an Anderson-Rubin (AR) test robust to weak identification. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5
Refugee Households: Heterogeneous Impacts by Baseline Market Participation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: Consumers (6 Month Lag)</i>								
Arrivals (6 Month Lag)	-0.081 (0.079)	0.056 (0.041)	0.258 (0.212)	0.061* (0.034)	0.082 (0.045)	0.048 (0.049)	-0.012 (0.033)	-0.017 (0.060)
Mean Dep. Var.	-0.26	0.85	3.79	0.26	0.59	0.85	0.51	0.82
Observations	1467	1510	1396	1510	1401	1398	1510	1493
R-squared	0.034	0.007	-0.024	-0.040	0.002	0.011	0.398	0.387
Weak ID (Wald F)	7.27	8.10	10.69	8.10	10.91	10.89	8.10	8.21
UnderID P-Value	0.028	0.029	0.023	0.029	0.023	0.023	0.029	0.028
Wild Boot. p-val	0.385	0.195	0.709	0.060	0.182	0.347	0.829	0.891
<i>Panel B: Consumers (12 Month Lag)</i>								
Arrivals (12 Month Lag)	-0.048 (0.045)	0.027 (0.020)	0.120 (0.111)	0.032* (0.020)	0.041 (0.022)	0.026 (0.023)	-0.008 (0.016)	-0.012 (0.029)
Mean Dep. Var.	-0.26	0.85	3.79	0.26	0.59	0.85	0.51	0.82
Observations	1467	1510	1396	1510	1401	1398	1510	1493
R-squared	0.007	0.059	0.008	-0.012	0.007	0.015	0.393	0.384
Weak ID (Wald F)	7.87	9.37	11.48	9.37	11.76	11.75	9.37	9.42
UnderID P-Value	0.035	0.037	0.036	0.037	0.035	0.035	0.037	0.037
Wild Boot. p-val	0.348	0.193	0.711	0.074	0.159	0.283	0.758	0.824
<i>Panel C: Producers (6 Month Lag)</i>								
Arrivals (6 Month Lag)	-0.102*** (0.072)	0.006 (0.019)	0.125* (0.089)	0.000 (0.021)	0.071 (0.066)	0.114 (0.099)	0.012 (0.033)	0.003 (0.053)
Mean Dep. Var.	-0.37	0.84	3.82	0.31	0.82	1.18	0.46	0.75
Observations	1498	1510	1397	1510	1398	1395	1510	1493
R-squared	0.039	0.180	0.022	0.091	0.033	0.023	0.582	0.541
Weak ID (Wald F)	3.64	3.73	3.72	3.73	3.74	3.74	3.73	3.83
UnderID P-Value	0.014	0.016	0.017	0.016	0.017	0.017	0.016	0.016
Wild Boot. p-val	0.001	0.728	0.061	0.986	0.356	0.296	0.875	0.981
<i>Panel D: Producers (12 Month Lag)</i>								
Arrivals (12 Month Lag)	-0.055*** (0.041)	0.003 (0.010)	0.058* (0.043)	0.001 (0.011)	0.037 (0.032)	0.060 (0.046)	0.005 (0.016)	-0.002 (0.027)
Mean Dep. Var.	-0.37	0.84	3.82	0.31	0.82	1.18	0.46	0.75
Observations	1498	1510	1397	1510	1398	1395	1510	1493
R-squared	0.006	0.180	0.011	0.091	0.024	0.012	0.583	0.542
Weak ID (Wald F)	3.41	3.78	4.08	3.78	4.06	4.10	3.78	3.93
UnderID P-Value	0.030	0.037	0.039	0.037	0.039	0.038	0.037	0.036
Wild Boot. p-val	0.000	0.758	0.085	0.960	0.308	0.269	0.910	0.974
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables correspond to household food security and various agricultural production outcomes. The sample is split into "Consumers" and "Producers" based on any market participation (agricultural earnings) in the first observation period. The independent variable is the change in refugee arrivals (in thousands) over the preceding 6 or 12 months. All models are estimated using a Shift-Share IV with year and household fixed effects. Standard errors in parentheses are clustered at the district level. P-values from a wild cluster bootstrap (using Webb weights) are reported to account for the small number of clusters. For Weak Identification Wald F-statistics below 10.00, the bootstrap p-values incorporate an Anderson-Rubin (AR) test robust to weak identification. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To identify the heterogeneous effects of arrivals across household types, I divide households based on their observed baseline market participation. Those that report positive agricultural earnings in their initial observation are classified as "producers", while those with no reported baseline agricultural earnings are classified as "consumers". Approximately 60% of host households and 41% of refugee households in the sample are designated as producers. Characteristics of the two household types can be seen in Table B.24. For this heterogeneous analysis, model specifications and estimation strategies reflect those in the previous section, though I relax the weak-instrument threshold to the standard rule-of-thumb of $F = 10$ (Staiger & Stock, 1997) due to a reduced sample size. For estimates below this threshold, I continue to employ the weak-instrument-robust Anderson-Rubin test.

5.3.1 Food Security

Tables 4 and 5 report estimates of the effects of arrivals on food security (CSI) and agricultural outcomes for host and refugee households, respectively, at the 6- and 12-month lags (Panels A–D). These findings indicate that the primary drivers of increased food insecurity among host households are those classified as consumers, while among refugee households, the impacts are concentrated among producers.

Host consumers experience significant negative outcomes. A 1,000-arrival increase at the district level is associated with a 0.0917 standard deviation decrease in food security at the 12-month lag. This corresponds to an increase of 1.697 CSI points per 1,000 arrivals respectively. When evaluated at the average district-level inflow over 12 months (9,182), these estimates imply a total increase of 15.58 CSI points for host consumer households. This would push the average host consumer household ($CSI = 18.51$) below the 14th group percentile (34 pts) of baseline food security measures, reflecting a large escalation in household distress.

Refugee producer households also experience significant deterioration in food security. A 1,000-arrival increase is associated with a 0.097 standard deviation decrease in food security at the 6-month lag and a 0.0525 standard deviation decrease at the 12-month lag. On the CSI scale, this translates to an increase of 2.64 and 1.43 CSI points per 1,000 arrivals, respectively. Using the average arrival volumes, this implies an approximate increase of 8.44 and 13.16 points at the 6- and 12-month marks. This magnitude would shift the average refugee producer household ($CSI = 27.3$) below the 28th and 22nd percentiles of food insecurity (35 and 40 points), respectively.

5.3.2 Market Participation

When looking at the extensive margin of overall market participation, I find that host-producer households are significantly more likely to report positive agricultural earnings. A 1,000-person increase over the previous 6- and 12-month periods is associated with a increase in the probability of reporting earnings by 7.92

percentage points and 3.71 percentage points, respectively. Aggregating to the average district level arrivals over the two periods (3,192 and 9,187), host producers are 25.3 and 34.1 percentage points more likely to report positive agricultural earnings—a substantial shift from the baseline mean of 25%.

Refugee consumer households exhibit similar trends, a 1,000 arrival increase leads to an increase in the probability of any earnings by 6.06 and 3.24 percentage points over the 6- and 12-month lags. At average arrival volumes, refugee consumer households are 19.37 and 29.75 percentage points more likely to report positive agricultural earnings. This again represents a large deviation from the group mean of 48%. Refugee consumer households are likely to be trading off the internal shadow-value of agricultural goods and the need for liquidity; they maintain household positioning with reliance on higher levels of in-kind (food) aid, increasing household market participation without altering total production. A refugee community leader in Imvepi noted this frequent use of direct food aid as a source of liquidity, "New arrivals receive food assistance, and you often see them in the market selling that food because they desperately need cash for other things... They can't just feed their families on beans and maize alone." (Participants in the Refugee Settlement Study, 2025).

5.3.3 Livestock Holdings

Livestock holdings display diverging patterns across household types. Among hosts, results show that the increase in livestock ownership, on both the extensive and intensive margins is largely driven by producer households. For these host-producer households, A 1,000-person increase in arrivals over the 6-month period raises the probability of owning any livestock by 5.5 percentage points and increases the total livestock holdings by 16.6 percent. Scaled to average 6-month arrivals (3,192 persons), host producers are 17.6 percentage points more likely to own livestock and increase herd size by approximately 53 percent. At the 12-month lag, scaled to average arrivals of 9,187 persons, host producers are 23.9 percentage points more likely to own livestock and increase herd size by approximately 73 percent. Among host consumer households, the 6-month lag shows a significant increase in the probability of any livestock ownership (6.6 percentage points), though this effect does not persist at the 12-month horizon.

In contrast, established refugee households—both consumers and producers—exhibit no significant response in livestock holdings across either time horizon or margin (Table 5, Columns 7 and 8). This highlights a further imbalance in adjustment capability: host households, particularly producers, possess sufficient initial assets to leverage livestock as a vehicle for wealth accumulation in response to arrival-driven demand shocks, while refugee households face binding capital and land constraints that disallow any potential gains.

5.3.4 Sales Volume and Harvest

Regarding the intensive margin of agricultural earnings, these findings do not show statistically significant effects of arrivals for any subgroup. However, there is an increase in total sales volume (kg) for host producers (Table 4) at the 6- and 12-month lags. Host producers increase the volume of sold production significantly with a 36.77 and 16.67 percent per thousand arrivals increase over the 6- and 12-month lags. This effect may be driving the increased market participation for hosts signaling a beneficial outcome driven by an increase in household profits, though the volume may not be large enough to statistically move the intensive margin of earnings.

Refugee producers, while exhibiting significant adverse food security impacts, also increase total harvest size in response to arrivals (Table 5, $p < 0.10$). On average, this group increases annual harvests by approximately 12.46 and 5.78 percent per thousand arrivals. Given the lack of corresponding increases in earnings or market participation and a larger reliance on cash-equivalent aid compared to refugee consumers, refugee producers appear to be increasing harvests and land-use intensity as a coping strategy to partially mitigate food insecurity driven by localized inflation. These households likely consume their increased harvests rather than selling it, substituting own-production for food goods that are no longer affordable with devalued vouchers. This mechanism was highlighted by a settlement official, who noted that recent reductions in food aid acted as a "major catalyst for increased agricultural production," forcing households who previously survived on aid to "rely on farming" (Participants in the Refugee Settlement Study, 2025).

5.3.5 Summarized Findings

The differentiated responses across household types align with their respective structural characteristics. Notably, while host consumers possess slightly higher average livestock holdings than producers (Table B.24), this is paired with significantly smaller crop production capabilities (nearly half of their producer counterparts), lower earnings and minimal access to aid. This combination leaves host consumers to experience pronounced decreases in food security without any gains in earnings, production, or market participation. By contrast, host producer households—the wealthiest and most productive group—appear better able to absorb price pressures and capitalize on demand shocks, as evidenced by their increased sales volume, market entry and persistent livestock accumulation.

Among refugee households, consumers have the smallest harvests, lowest earnings, and the least assets of any subgroup, along with high food insecurity. However, access to the highest levels of direct relief aid provides a buffer against negative price effects, allowing them to enter the market to sell small surpluses for liquidity without further worsening their comparatively low levels of food security. In contrast, refugee

producer households, despite having higher baseline market engagement and increased production, experience a significant increase in food insecurity. Refugee producer households are constrained by small, fixed plot allocations and low capital. This leads to an attempted mitigation of further food insecurity instead of market capitalization. Although refugee producers receive high amounts of aid, much of it is distributed in the form of vouchers that function as cash transfers. Local price inflation reduces the consumption value households receive when vouchers are traded for food goods in comparison to food relief aid, which scales with local price increases. These findings complement previous research (Mastrorillo et al., 2024) where rural highly food-insecure households with low market integration benefit from direct food aid over cash transfers.

These results underscore two central determinants of food security in the context of large-scale arrivals: secure market participation and the modality of aid received. Host producers, with higher assets and greater production capacity, are better able to offset the effects of high food prices by sustaining or expanding production and livestock ownership. Refugee consumers remain highly food insecure but partially insulated by access to direct food aid. Refugee producers, constrained by fixed land, limited assets and depreciating values of cash aid, experience worsening food security despite baseline market engagement and increased physical output. Finally, host consumer households, with low aid access and a reliance on purchased food, experience deterioration in food security due to price effects from arrivals. I argue the main mechanism driving these welfare effects is delayed but significant arrival-driven price inflation (see Section 7).

5.4 Robustness Checks

Before discussing the price mechanisms driving these welfare outcomes, I briefly assess the validity of the empirical strategy using placebo tests and sensitivity analyses (detailed fully in Appendix C). First, a temporal placebo using a forward-lead of arrivals and the shift-share instrument confirms that future arrivals largely do not predict current household outcomes, suggesting the estimates are not capturing pre-existing diverging trends. Second, a leave-one-district-out jackknife estimation provides evidence for the stability of the main estimates, reducing concerns of spatial outliers or localized effects from COVID-19. Finally, I note the stark divergence between the baseline OLS and IV estimates. OLS estimates generally indicate a weakly positive (negative) association between arrivals and food security (production outcomes). Rather than indicating an exclusion restriction violation, the sign reversal is consistent with the intentional placement of refugee arrivals into districts with greater baseline absorptive capacity (Participants in the Refugee Settlement Study, 2025)—a form of positive selection that is corrected by the instrument. Together, these exercises support the interpretation of the IV estimates as capturing the causal impact of refugee arrivals on household food security and agricultural production outcomes.

6 Crop Price Estimation

To isolate the causal impact of refugee arrivals on local food prices, I employ a dynamic panel model based on the Arellano-Bond GMM estimator (Arellano & Bond, 1991). The Arellano-Bond estimator is used to account for the dynamic nature of both refugee arrivals and local food prices which are reliant upon factors including previous seasonal pricing, productivity, storage capacity and local market characteristics. The estimator also allows for the ability to control for market level time-invariant characteristics while avoiding Nickell-Bias (Roodman, 2009) —a mechanical correlation between the lagged price and error term.

To ensure the model is well-specified, the primary estimation uses an expanded panel of 31 regional markets including both the 13 refugee-hosting settlements and 18 regional markets that do not host refugees (fully detailed in Table 6). This expansion of the group size (N) introduces a non-treated baseline to identify localized demand shocks induced by refugees, avoids the risk of instrument proliferation, and allows us to isolate the effect of arrivals from concurrent national inflation trends, such as the COVID-19 supply chain disruptions (D’Errico et al., 2024; Squarcina & Romano, 2024). For all non-hosting markets, local refugee arrivals are coded as zero.

The natural logarithm of the price of a given commodity is modeled as a function of its value in a previous period, recent refugee arrivals and a set of time-varying weather controls. The empirical model for commodity k in market s at time t is specified as:

$$\ln(P_{kst}) = \alpha_s + \lambda_{tk} + \rho \ln(P_{ks,t-1}) + \sum_{j=1}^3 \delta_j \cdot Arrivals_{s,t-j} + W'_{st}\gamma + \epsilon_{skt} \quad (10)$$

Where $\ln(P_{kst})$ is the natural logarithm of the real price of commodity k in market s in month t . α_s represents the unobserved, time-invariant market level characteristics. To capture time-varying commodity specific shocks common across markets, such as seasonal-price trends related to harvest timing, λ_{tk} will be used. Lagged price, $\ln(P_{ks,t-1})$, is included to construct an $AR(1)$ process. The coefficient ρ measures the degree of persistence in prices from month-to-month. $Arrivals_{s,t-j}$ represents the explanatory variables of focus, a temporal series of binned arrivals indicating the net inflow of refugees for the settlement market over multiple lagged periods (0-3 months, 3-6 months and 6-12 months from the current period). This allows the impact on prices to have a distributed lag effect. To directly account for seasonal weather variation at the settlement level, W_{st} will include the 6-month cumulative precipitation and maximum temperatures (along with their quadratics). Finally, the error term ϵ_{skt} is the idiosyncratic error term, clustered at the market level.

I estimate the model for three aid commodities (beans, maize, and oil) and two non-aid commodities (millet and sorghum). Beans and maize are produced locally and included in aid baskets, oil is aid-supported

but minimally produced locally, and millet and sorghum are staple foods predominantly consumed by South Sudanese refugees (WFP, 2023; Willis & Buck, 2007). This "basket of goods" allows us to measure a diverse range of outcomes including the temporal impact on prices from aid goods, the effects of good substitution depending on market availability, and the impacts of heterogeneous preferences due to cultural food norms.

6.1 Identification Strategy

Estimation with Equation 10 could yield biased results due in part to two main factors. First, the presence of the lagged price variable $\ln(P_{sk,t-1})$ would lead to a mechanical correlation with the market fixed effect α_s (Roodman, 2009), creating dynamic panel bias. Estimation bias may also arise from the endogeneity of the settlement level arrivals $Arrivals_{s,t-j}$; refugee arrivals are likely to be a function of and response to changes in food prices at the settlement level over time, generating reverse causality.

To address these endogeneity concerns, the Arellano-Bond system GMM estimator provides a two-part correction. First, the model is first-differenced to address market specific fixed effects α_s :

$$\Delta \ln(P_{kst}) = \rho \Delta \ln(P_{ks,t-1}) + \sum_{j=1}^3 \delta_j \cdot \Delta Arrivals_{s,t-j} + \Delta W'_{st} \gamma + \Delta \epsilon_{kst} \quad (11)$$

While this first-differencing removes the market fixed effect, it creates a new source of endogeneity with $\Delta \ln(P_{ks,t-1})$ now being correlated with $\Delta \epsilon_{kst}$.

This new source of endogeneity, along with the reverse causality concerns is addressed by Anderson and Hsiao (1982) and Arellano and Bond (1991) who introduce the use of lagged levels of the endogenous variables as instruments for the differenced equation 11, with $\ln(P_{ks,t-2})$, $\ln(P_{ks,t-3})$, etc., being used to instrument $\Delta \ln(P_{ks,t-1})$ under the assumption that the error term $\Delta \epsilon_{kst}$ is not correlated with the levels of the lagged instrumenting variables. For refugee arrivals, $\Delta Arrivals_{s,t-j}$ would be instrumented with $Arrivals_{s,t-j-n}$ for $n \geq 3$. The model treats previous arrivals as predetermined, assuming they are uncorrelated with the current error term. Unobserved shocks to food prices (ϵ_{kst}) may influence arrivals in the current period, but do not retroactively influence arrivals occurring in periods $t - 3$ or earlier.

To implement the Arellano-Bond GMM estimation, the endogenous lagged price and refugee arrival variables are instrumented using their own lags from $t - 3$ to $t - 5$. Weather and seasonal variables are treated as strictly exogenous and serve as their own instruments. To test the validity of the estimation strategy, the Arellano-Bond test for $AR(2)$ and Hansen Test of over-identification p-value are reported for each commodity. The Arellano-Bond test requires that the first-differenced errors $\Delta \epsilon_{kst}$ display first-order serial correlation $AR(1)$ but no second-order serial correlation $AR(2)$. A failure to reject the null hypothesis of not presenting the process of an $AR(2)$ model provides evidence for the estimation's specification and instrument validity.

The Hansen test of over-identifying restrictions assesses the overall validity of instruments by testing the null hypothesis that the full set of instruments is uncorrelated with the residuals. Failure to reject indicates a valid set of instruments.

7 Crop Price Results

This section presents the estimation results for crop specific price outcomes, focusing on the impact of refugee arrivals on local food prices. These results should be interpreted as a suggestive mechanism rather than direct causal evidence for the primary household findings due to limited temporal overlap (2021–2023) with the household data (2017–2021). Despite having only one year of direct overlap (2021), the underlying mechanisms are similar across both periods, with refugee arrivals increasing or maintaining persistently high levels throughout both observational windows (See Fig. 2 and 3). I also explore heterogeneity by predominant country of origin for settlement price outcomes and discuss robustness of the results when the sample is restricted strictly to hosting settlement markets.

Table 6: Price Effects of Refugee Arrivals

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: $\ln(\text{Price})$	Beans	Maize	Oil	Millet	Sorghum
$\ln(P_{t-1})$	0.9073*** (0.055 4)	0.9727*** (0.051 3)	0.9896*** (0.029 7)	0.5660*** (0.155 9)	1.0012*** (0.056 9)
$(\Delta)\text{Arrivals (0–3 mo)}$	−0.0013 (0.001 6)	−0.0087** (0.003 8)	0.0024 (0.001 9)	0.0017 (0.001 6)	−0.0058 (0.004 0)
$(\Delta)\text{Arrivals (3–6 mo)}$	−0.0053*** (0.001 5)	0.0064*** (0.002 3)	0.0006 (0.001 1)	−0.0028 (0.001 8)	0.0021 (0.003 0)
$(\Delta)\text{Arrivals (6–12 mo)}$	0.0111*** (0.002 2)	0.0081*** (0.002 1)	−0.0001 (0.001 1)	0.0026 (0.001 7)	0.0080*** (0.002 9)
Observations	856	834	834	275	781
Markets	32	32	32	13	31
AR(2) p-value	0.375	0.302	0.470	0.621	0.765
Hansen p-value	0.606	0.308	0.204	0.754	0.301
σ	0.122	0.344	0.091	0.117	0.159
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity in USD. All models are estimated using the Arellano-Bond GMM estimator. Standard errors, reported in parentheses, are clustered at the settlement level. The $(\Delta)\text{Arrivals}$ variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period.

Non-refugee settlement markets included: Busia, Fort Portal, Gulu, Hoima, Iganga, Jinja, Kapchorwa, Kasese, Lira, Makaratin, Masaka, Masindi, Mbale, Mbarara, Mukono, Owino, Soroti, Tororo, and Wakiso. Due to a lack of available market price data in these non-hosting regions, millet (Column 4) is estimated strictly on the original 13 refugee-hosting settlements.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6 reports the estimated impact of refugee arrivals on the natural logarithm of commodity prices, with standard errors clustered at the market level. The results show that the immediate impact (0-3 months) of arrivals on prices is statistically insignificant except for maize, which shows a small deflationary effect of 0.87% for every 1000 arrivals. These muted impacts may be the result of initial levels of food aid distributed to new arrivals being sufficient to meeting short-term needs without expanding market dependency. In the medium term (3–6 months), refugee arrivals are associated with a small deflationary effect of 0.5% per 1000 arrivals for beans, and a 0.81% *inflationary* effect for maize.

Notably, a pattern of long term inflation (6-12 months) emerges for both aid and non-aid goods grown locally, specifically beans and maize (Columns 1 & 2) and sorghum (Column 5). These results do not hold for oil (Column 3), also an aid good but minimally produced locally. There are no observable price increases due to refugee arrivals for millet (Column 4). These estimates suggest a 1,000-person increase in refugee arrivals is associated with a statistically significant price increase of approximately 1.1% for beans, 0.8% for maize and 0.8% for Sorghum. This pattern is consistent with local aid distribution policy (Brown & Torre, 2024). New arrivals initially receive full rations for the first three months, followed by a reduction in rations for the 3-6 month range, with a final reduction taking place after the 6-month post arrival mark. The aid distribution scheme likely pushes households to rely upon aid goods initially with market reliance and consumer preference being reflected in the long run, leading to inflationary price effects in the 6-12 month mark.

These delayed price effects are consistent with the delayed negative impact on household food security documented in Tables B.8 and B.9. Specifically, local commodity prices rise in the 6-12 months post arrival period, coinciding with the tapering of total aid distributed to arrivals. This induced demand for food goods is a likely mechanism for the adverse food security impacts for surrounding host and refugee households (Section 5).

The model’s distributed bins allow for the dynamic response of prices from refugee arrivals to be captured. The signs for the $AR(1)$ coefficient, weather controls and harvest season indicator all follow the expected outcome (see Table B.25). For all specifications, the Arellano-Bond test for $AR(2)$ fails to reject the null hypothesis of no second-order serial correlation, and the Hansen J-test fails to reject the null of joint instrument validity, providing evidence that the model is well-specified. Finally, the estimated standard deviation of the idiosyncratic error, σ , is reported for each model as an indicator of goodness-of-fit.

7.1 Heterogeneity by Refugee Origin

Table 7: Price Effects of Refugee Arrivals, Heterogeneity by Country of Origin

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: $\ln(\text{Price})$	Beans	Maize	Oil	Millet	Sorghum
$\ln(P_{t-1})$	0.9073*** (0.055 4)	0.9727*** (0.051 3)	0.9896*** (0.029 7)	0.4446*** (0.118 2)	0.9556*** (0.062 6)
(Δ)Arrivals (0–3 mo)	−0.0013 (0.001 6)	−0.0087** (0.003 8)	0.0024 (0.001 9)	0.0071 (0.005 3)	−0.0156 (0.011 7)
(Δ)Arrivals (3–6 mo)	−0.0053*** (0.001 5)	0.0064*** (0.002 3)	0.0006 (0.001 1)	0.0031 (0.003 6)	−0.0151* (0.007 7)
(Δ)Arrivals (6–12 mo)	0.0111*** (0.002 2)	0.0081*** (0.002 1)	−0.0001 (0.001 1)	0.0038 (0.003 4)	−0.0122 (0.009 6)
(Δ)Arr. (0–3 mo) $\times$ S. Sudan				−0.0058 (0.006 0)	0.0094 (0.012 8)
(Δ)Arr. (3–6 mo) $\times$ S. Sudan				−0.0093* (0.004 9)	0.0210** (0.008 2)
(Δ)Arr. (6–12 mo) $\times$ S. Sudan				0.0039 (0.004 3)	0.0230** (0.009 5)
S. Sudan Total Effect (0–3 mo) (<i>p-value</i>)				0.0013 (0.553)	−0.0062 (0.194)
S. Sudan Total Effect (3–6 mo) (<i>p-value</i>)				−0.0061 (0.100)	0.0059 (0.152)
S. Sudan Total Effect (6–12 mo) (<i>p-value</i>)				0.0077 (0.007)	0.0109 (0.000)
Observations	856	834	834	275	781
Settlements	32	32	32	13	31
AR(2) <i>p-value</i>	0.375	0.302	0.470	0.719	0.793
Hansen <i>p-value</i>	0.606	0.308	0.204	0.999	0.849
σ	0.122	0.344	0.091	0.118	0.157
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity in USD. All models are estimated using the Arellano-Bond GMM estimator. The variable “S. Sudan” is an indicator for settlements that are primarily South Sudanese refugees. Standard errors, reported in parentheses, are clustered at the settlement level. The (Δ)Arrivals variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period. To satisfy the exclusion restrictions and account for the altered error dynamics of the expanded panel, internal instruments are restricted to deeper lags ($t - 3$ to $t - 5$).

Non-refugee settlement markets included: Busia, Fort Portal, Gulu, Hoima, Iganga, Jinja, Kapchorwa, Kasese, Lira, Makaratin, Masaka, Masindi, Mbale, Mbarara, Mukono, Owino, Soroti, Tororo, and Wakiso. Due to a lack of available market price data in these non-hosting regions, millet (Column 4) is estimated on the original 13 refugee-hosting settlements.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

To explore heterogeneity in price effects by refugee origin, I interact refugee arrivals with an indicator for settlements that received predominantly South Sudanese refugees. This interaction is included for the estimation of millet and sorghum prices, due to these goods being preferred carbohydrate sources for refugees from South Sudan while not being preferentially consumed by refugees from the DRC. Sorghum is the dominant carbohydrate source for South Sudanese refugees, with millet consumption being much more limited (Willis & Buck, 2007).

The net effect and significance of arrivals in South Sudanese settlements is reported in Table 7. The interaction reveals that long-run inflation is stronger in South Sudanese settlements, reflecting cultural preferences. In these settlements, a 1,000-person increase in arrivals differentially raises millet prices by 0.77% and sorghum prices by 1.09% in the 6–12 month post-arrival window. By contrast, Congolese settlements exhibit a 1.51% ($p < 0.1$) decline in sorghum prices in the 3–6 month arrival window, with no detectable effects on millet prices. These findings are consistent with patterns of aid distribution and highlight the importance of cultural heterogeneity in consumption patterns for settlement communities.

Consistent with the standard pooled results (Table 6), the signs for the $AR(1)$ coefficient, weather controls and harvest season indicator all follow the expected signs (Table B.26). For all specifications, the Arellano-Bond test for $AR(2)$ fails to reject the null hypothesis of no second-order serial correlation, and the Hansen J-test fails to reject the null of joint instrument validity. However, the millet specification shows evidence of instrument proliferation, with an implausibly high Hansen p-value. This is driven by the limited market coverage for millet prices, which are recorded only in settlement markets in the time period, leading to a small group size relative to the number of instruments. The estimated standard deviations of the idiosyncratic error again follow a consistent pattern.

7.2 Robustness: Restricting to Settlement Markets

A potential concern with the expanded baseline estimation is that major regional markets (e.g., Gulu, Mbarara) possess fundamentally different market structures, storage capacities, and pricing dynamics compared to rural, insular refugee settlements. To ensure the observed inflationary effects are not an artifact of comparing rural settlements to urban centers, I restrict the panel exclusively to the 13 refugee-hosting settlements.

The restricted price analyses for both the pooled and heterogeneous estimations are reported in Tables C.3 and C.4. Despite the expected loss of statistical power from the restricted market analysis, the estimated price effects remain stable. The coefficients for the 6–12 month lagged arrivals remain positive, statistically

significant, and of a nearly identical magnitude for both beans and maize. Further, the heterogeneity analysis on the restricted panel further demonstrates the stability of the S. Sudan interaction terms.

It is important to note that in the restricted models, the group size ($N = 13$) introduces a risk of instrument proliferation, weakening the Hansen J-test and biasing results (Roodman, 2009). Though the pooled settlement estimation of Table C.3 remains stable, the restricted analysis by country of origin demonstrates evidence of instrument proliferation (Table C.4, Columns 4 and 5) with Hansen J-test p-values approaching 1.000 in the settlement-only estimations. While these point estimates align with the main specification, the inflated Hansen J-test p-values suggest these estimates should be interpreted with caution. The restricted heterogeneity analysis serves to confirm the direction and approximate magnitude of arrival effects hold within settlement-only estimations.

These results confirm that localized inflationary effects are a persistent feature of refugee arrivals within settlement communities and across goods, and not a byproduct of national/regional trends. Taken together, these price effects suggest that the initial allocation of aid goods to new arrivals temporarily delays demand-side market pressures, ultimately leading to long-term inflationary effects for both aid and non-aid commodities. I also find that refugee arrivals have heterogeneous impacts on local market prices depending on the country of origin. These price effects act as a central mechanism driving household consumption and agricultural production outcomes for both host and refugee communities, as discussed previously in Sections 5.1 and 5.2.

8 Conclusion

This paper demonstrates the causal impacts of refugee arrivals on settlement-level commodity prices, as well as the long-term food security and agricultural earnings of host and established refugee households across 11 districts in Uganda, the largest refugee-hosting country in Africa.

The findings suggest that refugee arrivals lead to statistically significant and negative impacts on food security for both host and established refugee households. The estimates when scaled to average annualized district-level arrivals (9,182), demonstrate that hosts experience a 9.1-point CSI increase (a 0.55 standard deviation shock), and refugees an 11.4-point CSI increase (a 0.44 standard deviation shock). This pushes both groups' average households into the bottom quartile of baseline food security. Despite these adverse consumption outcomes, I observe a large expansion in agricultural market participation. Scaled to average inflows, host and established refugee households are 26.6 and 11.2 percentage points more likely to report positive agricultural earnings, respectively. Notably, these market shifts are accompanied by diverging asset accumulation: while host households significantly increase their livestock holdings as a form of wealth accumulation, refugees exhibit no such behaviors, underscoring binding capital and land constraints.

Substantial evidence indicates the primary mechanism driving these outcomes is localized price inflation due to refugee arrivals. I find that arrivals lead to delayed but significant inflation for both aid and non-aid food commodities grown within settlements. These inflationary effects emerge in the 6- to 12-month post-arrival window, which is consistent with aid policies that gradually reduce in-kind assistance, increasing refugees' market dependence over time. The resulting price inflation erodes household purchasing power, generating the observed negative food security outcomes while simultaneously encouraging increased market activity to maintain liquidity.

Deterioration in food security is driven primarily by host consumer and established refugee producer households, though through differing mechanisms. Host consumers lack agricultural assets and aid buffers, leaving them highly exposed to rising food prices. Refugee producers, despite successfully increasing their physical harvests, suffer because their reliance on cash-equivalent vouchers leaves them highly vulnerable to inflation; their purchasing power drops, forcing them to consume their surplus production rather than sell it.

These price effects are not uniform across the population. Subgroup analysis based on baseline market participation reveals that outcomes are heavily dictated by access to capital and aid modality. Host consumers and refugee producer households are uniquely exposed to price effects due to low levels of inflation-resistant in-kind aid and a reduced ability to capitalize on demand shocks. I also find evidence suggesting increased market participation among host producer and refugee consumer households in response to arrivals. These groups respond similarly but differ in the respective mechanism. Host producers, possessing high levels of baseline assets, capitalize on refugee inflows with increased agricultural sales volume, market participation, and livestock accumulation, avoiding adverse food security impacts. In contrast, refugee consumers—who receive the highest levels of inflation-resistant direct food aid—are partially insulated from price shocks. They likely rely on higher levels of food aid to improve household liquidity while not further worsening high levels of baseline food insecurity, reflecting the need in settlements to sustain cash-on-hand.

There are some limitations to this analysis. Measurement challenges arise from the alignment of data sources: climate indicators representing rolling six-month windows may not fully capture Uganda's multiple harvest cycles or staggered survey timing, while price data (2021–2023) overlap only partially with the final wave of the household panel (2017–2021). Consequently, the price analysis should be viewed as supportive evidence of the proposed mechanism rather than direct identification. While anticipatory effects cannot be entirely ruled out, the temporal placebo test (See Table C.1) provides reassurance that the main estimates are not driven by household or settlement level expectations of future arrivals.

The outcomes identified offer relevant information for policymakers. The persistent food security challenges faced by host and refugee communities suggest that Uganda's progressive model may not entirely buffer negative impacts from arrivals. It is important to give context to these numbers. While the estimated

impacts on food security are significant, the corresponding increases in market participation imply that the Ugandan model partially shields receiving communities from more significant impacts in light of historic regional instability. Policies could focus on shoring up the channels that would mitigate the adverse effects of localized price inflation. This reinforces the importance of increasing targeted direct food aid over aid in the form of vouchers, improving long-term support in the agricultural sector and enhancement of external market access. Taking into consideration the decreases in access to aid and increasing uncertainty regarding funding for international aid agencies (UNHCR, 2025), this may be a challenging recommendation.

The conclusions drawn in this paper motivate the exploration of further paths of understanding Uganda's refugee policy. Investigating heterogeneous impacts across intensity of conflict periods and camps is important for understanding the distributional consequences of large inflows. Further exploration of price effects for a wider range of commodities is warranted. Evaluating commodity level price volatility and different aid modalities across settlements may improve the understanding of the efficacy of assistance in a wide-range of market contexts. Finally, integration of satellite and higher resolution climate data could allow for analysis on specific crop choices and the impacts on land use decision-making associated with refugee arrivals and changing market conditions.

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A Theoretical Model Appendix

I analyze a local food and labor market in refugee settlements with no external market access, following the agricultural household framework of Singh et al. (1986). The model considers host households, established refugees, and new arrivals, and incorporates three goods: food (X_a), a numeraire non-food good (X_m), and leisure (X_l). Production occurs for hosts and established refugees ($Q^i = L_i^\alpha \bar{A}_i^\beta$), while new arrivals do not produce any staple crop. Aid consists of direct food transfers (Z_a) and cash (Z_m), with hosts receiving the least aid but possessing the best land. Refugee households are able to sell labor at S_i , while Hosts can only sell to themselves internalizing the $F_H = S_H$.

I assume $\alpha + \beta = 1$ for the Cobb-Douglas production technology and $\gamma + \delta + \theta = 1$ for the Cobb-Douglas utility function parameters.

A.1 Host Households (H)

Hosts produce using both family labor (F_H) and hired labor (H_H):

$$L_H = F_H + H_H, \quad Q_H = L_H^\alpha \bar{A}_H^\beta \quad (\text{A.1})$$

Their profit function is:

$$\pi_H = p_a L_H^\alpha \bar{A}_H^\beta - w L_H \quad (\text{A.2})$$

with first-order condition for labor demand:

$$\frac{d\pi_H}{dL_H} = \alpha p_a L_H^{\alpha-1} \bar{A}_H^\beta - w = 0 \implies L_H^* = \left(\frac{\alpha p_a}{w} \right)^{\frac{1}{\beta}} \bar{A}_H \quad (\text{A.3})$$

and optimal production and profit:

$$Q_H^* = \left(\frac{\alpha p_a}{w} \right)^{\frac{\alpha}{\beta}} \bar{A}_H, \quad \pi_H^* = p_a \beta \bar{A}_H \left(\frac{p_a \alpha}{w} \right)^{\frac{\alpha}{\beta}} \quad (\text{A.4})$$

Hosts maximize utility:

$$U^H = \gamma \ln X_a + \delta \ln X_m + \theta \ln X_l \quad (\text{A.5})$$

subject to their budget constraint including aid:

$$p_a X_a + X_m + w X_l = w T + \pi_H^* + Z_m + p_a Z_a \quad (\text{A.6})$$

Solving yields:

$$X_a^{*H} = \gamma \left[\frac{wT + \pi_H^* + Z_m}{p_a} + Z_a \right] \quad (\text{A.7})$$

$$X_m^{*H} = \delta [wT + \pi_H^* + Z_m + p_a Z_a] \quad (\text{A.8})$$

$$X_l^{*H} = \theta \left(T + \frac{\pi_H^* + Z_m + p_a Z_a}{w} \right) \quad (\text{A.9})$$

$$F_H^* = T(1 - \theta) - \theta \left[\frac{\pi_H^* + Z_m + p_a Z_a}{w} \right] \quad (\text{A.10})$$

$$H_H^* = \left(\frac{\alpha p_a}{w} \right)^{\frac{1}{\beta}} \bar{A}_H - \left[T(1 - \theta) - \theta \left[\frac{\pi_H^* + Z_m + p_a Z_a}{w} \right] \right] \quad (\text{A.11})$$

A.2 Established Refugees (R)

Established refugees cannot hire labor, making their problem non-separable, but they can also sell labor off-farm S_R . They maximize:

$$U^R = \gamma \ln X_a + \delta \ln X_m + \theta \ln(T - F_R - S_R) \quad (\text{A.12})$$

subject to the cash income constraint:

$$p_a X_a + X_m = p_a F_R^\alpha \bar{A}_R^\beta + w S_R + Z_m + p_a Z_a \quad (\text{A.13})$$

The Lagrangian is:

$$\mathcal{L} = \gamma \ln X_a + \delta \ln X_m + \theta \ln(T - F_R - S_R) + \lambda(p_a F_R^\alpha \bar{A}_R^\beta + w S_R + Z_m + p_a Z_a - p_a X_a - X_m) \quad (\text{A.14})$$

Optimal consumption is:

$$X_a^{*R} = \gamma \left[\beta \bar{A}_R \left(\frac{\alpha p_a}{w} \right)^{\frac{\alpha}{\beta}} + Z_a + \frac{Z_m + T}{p_a} \right] \quad (\text{A.15})$$

$$X_m^{*R} = \delta \left[p_a \beta \bar{A}_R \left(\frac{\alpha p_a}{w} \right)^{\frac{\alpha}{\beta}} + p_a Z_a + Z_m + wT \right] \quad (\text{A.16})$$

$$F_R^* = \left(\frac{\alpha p_a}{w} \right)^{\frac{1}{\beta}} \bar{A}_R \quad (\text{A.17})$$

$$S_R^* = T - \left(\frac{\alpha p_a}{w} \right)^{\frac{1}{\beta}} \bar{A}_R - \frac{\theta p_a}{w} \cdot \left[\beta \bar{A}_R \left(\frac{\alpha p_a}{w} \right)^{\frac{\alpha}{\beta}} + Z_a + \frac{Z_m + T}{p_a} \right] \quad (\text{A.18})$$

A.3 New Arrival Households (N)

New arrivals do not produce, they maximize:

$$U^N = \gamma \ln X_a + \delta \ln X_m + \theta \ln(T - S_N) \quad (\text{A.19})$$

Subject to the budget:

$$p_a X_a + X_m + w(T - S_N) = wT + Z_m + p_a Z_a \quad (\text{A.20})$$

Optimal consumption is:

$$X_a^{*N} = \gamma \left[\frac{wT + Z_m}{p_a} + Z_a \right] \quad (\text{A.21})$$

$$X_m^{*N} = \delta [wT + Z_m + p_a Z_a] = \delta m_N \quad (\text{A.22})$$

$$S_N^* = T(1 - \theta) - \theta \left[\frac{Z_m + p_a Z_a}{w} \right] \quad (\text{A.23})$$

A.4 Market Clearing

Let N_H, N_R, N_N denote the number of households of each type. Equilibrium prices (p_a^*, w^*) satisfy the clearing of the food and labor markets. Let $\phi = (\frac{\alpha p_a}{w})^{\frac{1}{\beta}}$.

Food Market: Total food demand equals total production and aid.

$$\begin{aligned} \sum_{i \in \{H, R, N\}} N_i X_a^{*i} &= \sum_{i \in \{H, R\}} N_i Q_a^{*i} + \sum_{i \in \{H, R, N\}} N_i Z_a \\ F(p_a, w, N_i, \bar{A}_i, Z_i) &\equiv N_H \gamma \left[\beta \bar{A}_H \phi^\alpha + Z_a^H + \frac{Z_m^H + T}{p_a} \right] \\ &\quad + N_R \gamma \left[\beta \bar{A}_R \phi^\alpha + Z_a^R + \frac{Z_m^R + T}{p_a} \right] \\ &\quad + N_N \gamma \left[\frac{wT + Z_m^N}{p_a} + Z_a^N \right] \\ &\quad - N_H \bar{A}_H \phi^\alpha - N_R \bar{A}_R \phi^\alpha - Z_a^H N_H - Z_a^R N_R - Z_a^N N_N = 0 \end{aligned}$$

Labor Market: Total labor inputs equals total labor supply.

$$\text{Demand} = N_H H_H^* + N_H F_H^* + N_R F_R^*$$

$$\text{Supply} = N_H F_H^* + N_R F_R^* + N_R S_R^* + N_N S_N^*$$

$$\implies N_H H_H^* = N_R S_R^* + N_N S_N^*$$

$$\begin{aligned} L(p_a, w, \bar{A}_i, Z_i) \equiv & N_H \left[\phi \bar{A}_H - \left[T(1 - \theta) - \theta \left[\frac{p_a \beta \bar{A}_H \phi^\alpha + Z_m^H + p_a Z_a^H}{w} \right] \right] \right] \\ & - N_R \left(T - \phi \bar{A}_R - \frac{\theta p_a}{w} \left[\beta \bar{A}_R \phi^\alpha + Z_a^R + \frac{Z_m^R + T}{p_a} \right] \right) \\ & - N_N \left[T(1 - \theta) - \theta \left[\frac{Z_m^N + p_a Z_a^N}{w} \right] \right] = 0 \end{aligned}$$

A.5 Comparative Statics: Response to New Arrivals

There are two equilibrium conditions

$$F(p_a, w, X) = 0, \quad L(p_a, w, X) = 0,$$

with X containing N_N (and other exogenous variables $\bar{A}_i, Z_a^i, Z_m^i$). The Jacobian matrix is

$$J = \begin{pmatrix} F_{p_a} & F_w \\ L_{p_a} & L_w \end{pmatrix}.$$

Implicit differentiation with respect to N_N yields

$$\begin{pmatrix} F_{p_a} & F_w \\ L_{p_a} & L_w \end{pmatrix} \begin{pmatrix} dp_a \\ dw \end{pmatrix} = - \begin{pmatrix} F_{N_N} \\ L_{N_N} \end{pmatrix} dN_N.$$

Therefore

$$\begin{pmatrix} dp_a \\ dw \end{pmatrix} = -J^{-1} \begin{pmatrix} F_{N_N} \\ L_{N_N} \end{pmatrix} dN_N.$$

With the 2×2 inverse, closed-form expressions are:

$$\boxed{dp_a = -\frac{L_w F_{N_N} - F_w L_{N_N}}{\det J} dN_N, \quad dw = -\frac{-L_{p_a} F_{N_N} + F_{p_a} L_{N_N}}{\det J} dN_N}$$

where

$$\det J = F_{p_a} L_w - F_w L_{p_a}$$

The partial derivatives of the market clearing functions $F(\cdot)$ and $L(\cdot)$ are:

$$\begin{aligned}
F_{p_a} &= -\frac{(\bar{A}_R N_R + \bar{A}_H N_H) \alpha p_a \left(\frac{\alpha p_a}{w}\right)^{\frac{\alpha}{\beta}} + (N_N T w + N_N Z_m^N) \beta \gamma}{\beta p_a^2}, \\
F_w &= \frac{N_N T \beta \gamma w + (\bar{A}_R N_R + \bar{A}_H N_H) p_a \alpha \left(\frac{p_a \alpha}{w}\right)^{\frac{\alpha}{\beta}}}{p_a \beta w}, \\
L_{p_a} &= \frac{(\bar{A}_R N_R \beta^2 + \bar{A}_R N_R \alpha \beta) \theta p_a \left(\frac{\alpha p_a}{w}\right)^{\frac{\alpha}{\beta}} + (\bar{A}_R N_R + \bar{A}_H N_H) w \left(\frac{\alpha p_a}{w}\right)^{\frac{1}{\beta}} + N_R Z_a^R \beta \theta p_a}{w \beta p_a}, \\
L_w &= -\frac{(\bar{A}_R N_R + \bar{A}_H N_H) \left(\frac{p_a \alpha}{w}\right)^{\frac{1}{\beta}} w + (\bar{A}_R N_R p_a \beta^2 + \bar{A}_R N_R p_a \alpha \beta) \theta \left(\frac{p_a \alpha}{w}\right)^{\frac{\alpha}{\beta}} + (N_R Z_a^R p_a + N_R Z_m^R + N_R T) \beta \theta}{\beta w^2}, \\
F_{N_N} &= \gamma \left(\frac{T w + Z_m^N}{p_a} + Z_a^N \right) - Z_a^N, \\
L_{N_N} &= -\left[T(1 - \theta) - \theta \left[\frac{Z_m^N + p_a Z_a^N}{w} \right] \right] = (T + 1)\theta - T.
\end{aligned}$$

The explicit comparative static derivatives are

$$\frac{dp_a}{dN_N} = -\frac{L_w \left[\gamma \left(\frac{T w + Z_m^N}{p_a} + Z_a^N \right) - Z_a^N \right] - F_w [(T + 1)\theta - T]}{F_{p_a} L_w - F_w L_{p_a}}. \quad (\text{A.24})$$

I expect the denominator to be positive due to Walrasian stability $\det J > 0$. Therefore, the numerator determines the sign. L_w , the change in labor demand from wage fluctuations, is functionally negative. F_{N_N} is the induced food demand from refugee arrivals. The sign is going to be determined by the magnitude and direction of food demanded $\gamma \left(\frac{T w + Z_m^N}{p_a} + Z_a^N \right)$ offset by the aid allocated to new arrivals $-Z_a^N$. If aid is unit allocated to consumption, there would be no expected effects on net demand. However, with a relative constraint in food aid availability, I expect F_{N_N} to be a positive increase in food demanded. F_w is mechanically positive, as higher wages constrain supply. L_{N_N} is the new-arrivals contribution to market labor; unless θ , the preference for leisure, is close to one, then arrivals will contribute to the labor supply, making L_{N_N} negative: $(T + 1)\theta - T < 0$. Overall, unless the wage effects on production outweigh the induced demand, there is likely to be food price inflation from arrivals at the settlement level. $\frac{dp_a}{dN_N} > 0$

$$\frac{dw}{dN_N} = -\frac{-L_{p_a} \left[\gamma \left(\frac{T w + Z_m^N}{p_a} + Z_a^N \right) - Z_a^N \right] + F_{p_a} [(T + 1)\theta - T]}{F_{p_a} L_w - F_w L_{p_a}}. \quad (\text{A.25})$$

Again, the denominator is expected to be positive ($\det J > 0$). The term L_{p_a} captures labor supply responses to food price changes. It is functionally positive, due to a higher food price raising income from production leading to an increased marginal return to labor inputs. F_{N_N} is the likely positive net food demand from new arrivals (as shown above). F_{p_a} is mechanically negative, reflecting the standard inverse relationship between consumption and price. L_{N_N} is the new-arrivals contribution to market labor, which likely decreases wages. Overall, this combines two aspects from new arrivals on wages. The first is the positive price effect on labor demand $-L_{p_a}F_{N_N}$, which will push wage increases when labor demand rises. The second is the labor supply effect $F_{p_a}L_{N_N}$, which is likely to decrease wages with new arrivals entering the labor market. Unless the food-price induced labor demand is greater than the effects of the new labor supply, there is likely be wage deflation from arrivals at the settlement level. $\frac{dw}{dN_N} < 0$.

A.5.1 Testable Hypotheses from Comparative Statics

The comparative static results in the equations for dp_a/dN_N and dw/dN_N lead directly to relevant and testable hypotheses. First, an increase in new-arrivals ΔN_N will increase local crop prices p_a . This occurs due to demand for additional consumption $F_{N_N} > 0$ dominating wage effects. The price effect is likely to be smaller, and potentially negative, when new arrivals receive sufficiently large amounts of in-kind food aid Z_a^N . Another avenue for price effects to be attenuated is through household production decisions. With localized commodity inflation and a reduction in wages, hosts and established refugees may expand production based on initial land allocations and productivity $\bar{A}_i$. The wage-depressing effect is lessened if new arrivals receive significant cash aid (Z_m^N). This raises their demand for leisure, reducing their market labor supply (S_N^*) and making the L_{N_N} shock smaller. Finally, there are likely to be heterogeneous welfare impacts. Hosts (H) may gain from an influx if the resulting price increases raise their farm profits (π_H^*) sufficiently. Established Refugees (R) and New Arrivals (N) are likely to lose if p_a rises, as this reduces the real purchasing power of their cash aid (Z_m^i) and labor income.

B Additional Tables and Figures

Table B.1: Sample Observations by District and Household Type Over Time

District	2017	2018	2019	2020	2021	Total
<i>Panel A: Host Households</i>						
Adjumani	153	0	104	125	130	512
Arua	395	0	426	448	479	1,748
Isingiro	0	0	675	312	374	1,361
Kamwenge	0	151	121	135	132	539
Kikuube	0	0	285	121	116	522
Kiryandongo	199	0	84	142	183	608
Koboko	0	0	23	21	24	68
Kyegegwa	0	153	353	218	241	965
Lamwo	158	0	308	174	285	925
Moyo	163	0	371	159	296	989
Yumbe	239	0	99	193	198	729
Total	1,307	304	2,849	2,048	2,458	8,966
<i>Panel B: Refugee Households</i>						
Adjumani	198	0	164	155	158	675
Arua	476	0	587	271	654	1,988
Isingiro	0	0	581	224	330	1,135
Kamwenge	0	181	144	150	159	634
Kikuube	0	0	269	124	111	504
Kiryandongo	318	0	153	214	287	972
Koboko	0	0	151	109	150	410
Kyegegwa	0	197	405	232	286	1,120
Lamwo	200	0	391	270	345	1,206
Moyo	198	0	505	81	316	1,100
Yumbe	284	0	193	204	275	956
Total	1,674	378	3,543	2,034	3,071	10,700

Notes: The table reports the distribution of sample observations by district and survey year. Panel A displays host household observations, and Panel B displays refugee household observations.

Table B.2: Summary Statistics: Refugee vs Host Households - Full Dataset

	Refugees	Hosts	Normalized Difference
<i>Panel A: Outcomes</i>			
Coping Strategies Index	24.04	16.43	0.28
Any Harvest	0.87	0.96	-0.24
Harvest (KG)	164.54	880.78	-0.51
Any Earnings	0.35	0.61	-0.38
Ag Earnings (USD)	10.27	90.08	-0.38
Total KG Crops Sold	39.13	270.17	-0.38
Any Livestock	0.42	0.69	-0.40
Livestock (Heads)	1.91	3.79	-0.30
<i>Panel B: Household Characteristics</i>			
Years of Schooling	8.28	7.99	0.04
Household Size	6.01	6.46	-0.10
Total Land (Acres)	0.51	4.23	-0.67
Normalized Asset Index	-0.34	0.41	-0.57
Voucher Aid (USD)	142.52	6.91	0.58
Relief Aid (USD)	139.90	1.94	0.64
Months in Settlement	204.18	.	.
Observations	10,702	8,975	.

Notes: The table reports sample means for the full balanced panel (all waves). The normalized difference is calculated as the difference in means divided by the square root of the average of the group variances.

Table B.3: Refugee Household Characteristics by Number of Observations

	1	2	3	4
<i>Panel A: Outcomes</i>				
Coping Strategies Index	22.98	22.66	22.61	25.95
Any Harvest	0.92	0.84	0.85	0.86
Harvest (KG)	113.24	127.88	152.06	201.56
Any Earnings	0.45	0.33	0.31	0.34
Ag Earnings (USD)	6.82	7.00	9.62	12.88
Total KG Crops Sold	24.19	23.62	34.21	52.57
Any Livestock	0.43	0.36	0.43	0.43
Livestock (Heads)	2.10	1.46	1.91	2.01
<i>Panel B: Household Characteristics & Aid</i>				
Years of Schooling	9.03	7.90	8.20	8.21
Household Size	5.23	5.78	6.05	6.34
Total Land (Acres)	0.46	0.44	0.50	0.56
Normalized Asset Index	-0.38	-0.40	-0.28	-0.34
Voucher Aid (USD)	84.60	112.06	140.52	169.52
Relief Aid (USD)	146.60	140.86	153.65	127.04
Months in Settlement	186.18	249.07	194.96	201.15
Observations	1,529	1,585	3,321	4,267

Notes: The table reports sample means for refugee households by the number of observations present in the panel.

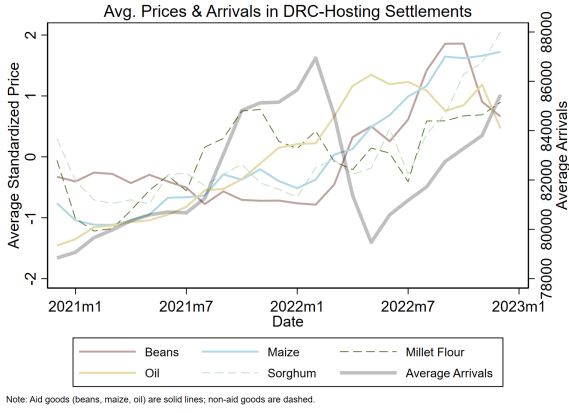
Table B.4: Summary Statistics of Commodity Prices by Settlement Type

Settlement Type	Price in U.S. Dollars per Kilogram				
	Beans	Maize	Millet Flour	Oil	Sorghum
Primarily Congolese	0.58 (0.19)	0.33 (0.15)	1.00 (0.12)	2.44 (0.45)	0.48 (0.14)
Primarily South Sudanese	0.83 (0.20)	0.39 (0.13)	1.13 (0.22)	2.19 (0.40)	0.41 (0.13)

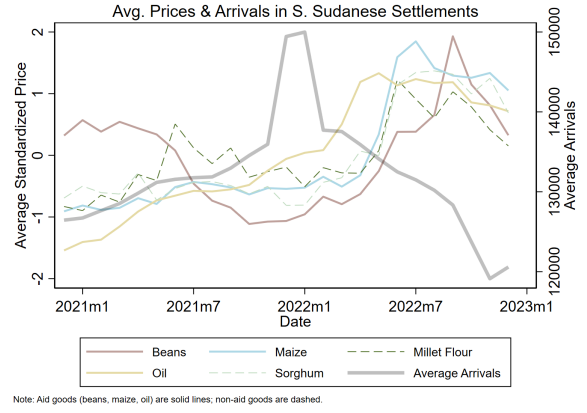
Notes: This table reports the mean commodity prices, with standard deviations in parentheses. Prices are aggregated for settlements hosting primarily Congolese refugees versus those hosting primarily South Sudanese refugees.

Table B.5: Average Commodity Prices by Settlement (Mean and SD)

Settlement	Beans	Maize	Millet Flour	Oil	Sorghum
Adjumani	0.93 (0.19)	0.43 (0.14)	1.09 (0.22)	2.32 (0.32)	0.41 (0.11)
Bidibidi	0.87 (0.09)	0.37 (0.10)	1.16 (0.05)	1.94 (0.38)	0.32 (0.08)
Imvepi	0.71 (0.23)	0.37 (0.12)	1.21 (0.23)	2.10 (0.50)	0.49 (0.15)
Kiryandongo	0.84 (0.16)	0.34 (0.13)	1.27 (0.20)	2.27 (0.37)	0.51 (0.12)
Kyaka	0.51 (0.15)	0.32 (0.13)	1.06 (0.10)	2.46 (0.41)	0.52 (0.08)
Kyangwali	0.60 (0.19)	0.29 (0.13)	1.01 (0.13)	2.64 (0.51)	0.56 (0.15)
Lobule	0.77 (0.11)	0.38 (0.13)	0.93 (0.12)	2.22 (0.41)	0.51 (0.13)
Nakivale	0.55 (0.18)	0.34 (0.16)	0.96 (0.11)	2.38 (0.44)	0.41 (0.12)
Oruchinga	0.52 (0.16)	0.34 (0.17)	0.99 (0.14)	2.44 (0.40)	0.41 (0.17)
Palabek	0.96 (0.18)	0.42 (0.14)	0.92 (0.16)	2.36 (0.40)	0.37 (0.15)
Palorinya	0.80 (0.17)	0.36 (0.12)	1.26 (0.19)	2.17 (0.33)	0.38 (0.12)
Rhino	0.72 (0.26)	0.44 (0.14)	0.94 (0.12)	2.15 (0.33)	0.39 (0.08)
Rwamwanja	0.50 (0.17)	0.31 (0.15)	1.06 (0.05)	2.52 (0.47)	0.48 (0.11)



(a) DRC-hosting Settlements



(b) S. Sudanese Settlements

Figure B.1: Comparison of price and arrival dynamics across settlement types. The left panel shows settlements hosting refugees from DRC, while the right panel shows settlements hosting refugees from South Sudan.

Table B.6
Hosts: First-Stage Estimates of Shift-Share Instrument on Refugee Arrivals

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: 3 Month Lag</i>								
Instrument	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
Observations	3394	3367	3254	3367	3243	3254	3367	3312
R-squared	0.639	0.635	0.639	0.635	0.640	0.638	0.635	0.631
F-Stat (First Stage)	10.01	9.43	9.23	9.43	9.57	9.52	9.43	9.13
Wild Boot. p-val	0.018	0.019	0.020	0.019	0.018	0.018	0.019	0.019
<i>Panel B: 6 Month Lag</i>								
Instrument	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.001)
Observations	3394	3367	3254	3367	3243	3254	3367	3312
R-squared	0.725	0.724	0.725	0.724	0.724	0.724	0.724	0.723
F-Stat (First Stage)	5.69	5.80	5.60	5.80	5.64	5.72	5.80	6.00
Wild Boot. p-val	0.025	0.023	0.026	0.023	0.025	0.024	0.023	0.023
<i>Panel C: 12 Month Lag</i>								
Instrument	-0.003*** (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
Observations	3394	3367	3254	3367	3243	3254	3367	3312
R-squared	0.792	0.792	0.792	0.792	0.789	0.792	0.792	0.791
F-Stat (First Stage)	10.67	10.68	10.40	10.68	10.32	10.93	10.68	11.06
Wild Boot. p-val	0.012	0.012	0.012	0.012	0.012	0.010	0.012	0.011
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the first-stage estimates for the Shift-Share IV specification. The dependent variable in all columns is the change in refugee arrivals (in thousands). The instrument is the shift-share construction described in the text. Standard errors in parentheses are clustered at the district level. Wild cluster bootstrap p-values are reported to account for the small number of clusters. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.7
Refugees: First-Stage Estimates of Shift-Share Instrument on Refugee Arrivals
(Refugee Household Sample)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: 3 Month Lag</i>								
Instrument	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Observations	2990	3037	2834	3037	2838	2833	3037	3006
R-squared	0.581	0.584	0.577	0.584	0.580	0.580	0.584	0.584
F-Stat (First Stage)	19.13	18.46	20.56	18.46	20.33	20.56	18.46	18.75
Wild Boot. p-val	0.009	0.009	0.005	0.009	0.005	0.005	0.009	0.009
<i>Panel B: 6 Month Lag</i>								
Instrument	-0.002** (0.001)	-0.002** (0.001)	-0.002** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.002** (0.001)	-0.002** (0.001)
Observations	2990	3037	2834	3037	2838	2833	3037	3006
R-squared	0.667	0.672	0.672	0.672	0.674	0.673	0.672	0.672
F-Stat (First Stage)	6.80	7.51	7.85	7.51	7.98	7.86	7.51	7.64
Wild Boot. p-val	0.037	0.039	0.035	0.039	0.034	0.035	0.039	0.038
<i>Panel C: 12 Month Lag</i>								
Instrument	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
Observations	2990	3037	2834	3037	2838	2833	3037	3006
R-squared	0.739	0.742	0.723	0.742	0.725	0.724	0.742	0.740
F-Stat (First Stage)	11.96	16.06	17.02	16.06	17.23	17.03	16.06	16.48
Wild Boot. p-val	0.011	0.014	0.012	0.014	0.011	0.012	0.014	0.013
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the first-stage estimates for the Shift-Share IV specification. The dependent variable in all columns is the change in refugee arrivals (in thousands). The instrument is the shift-share construction described in the text. Standard errors in parentheses are clustered at the district level. Wild cluster bootstrap p-values are reported to account for the small number of clusters. *Significance levels:* * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8
Host Households: Effect of Refugee Arrivals on Food Security

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	0.0319* (0.0190)	0.0377* (0.0221)	0.0526* (0.0278)	0.0539** (0.0265)	0.0539 (0.0710)	0.1279** (0.0593)	0.1279 (0.0818)
Observations	3394	3394	3394	3394	3394	3386	3386
R-squared	0.023	0.024	0.063	0.031	0.031	0.027	0.027
UnderID (LM Test)						430.798	4.481
UnderID P-Value						0.000	0.034
Weak ID (Wald F-Stat)						531.820	10.011
Wild Boot. p-val					0.538		0.285
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	0.0187*** (0.0063)	0.0272*** (0.0087)	0.0088 (0.0112)	0.0076 (0.0107)	0.0076 (0.0240)	-0.1273*** (0.0344)	-0.1273** (0.0631)
Observations	3394	3394	3394	3394	3394	3386	3386
R-squared	0.025	0.026	0.062	0.029	0.029	-0.048	-0.048
UnderID (LM Test)						209.070	5.793
UnderID P-Value						0.000	0.016
Weak ID (Wald F-Stat)						229.653	5.691
Wild Boot. p-val					0.802		0.043
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	0.0102*** (0.0025)	0.0150*** (0.0035)	0.0097* (0.0053)	0.0089* (0.0049)	0.0089 (0.0101)	-0.0597*** (0.0162)	-0.0597** (0.0273)
Observations	3394	3394	3394	3394	3394	3386	3386
R-squared	0.027	0.029	0.063	0.030	0.030	-0.054	-0.054
UnderID (LM Test)						224.359	6.397
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						248.334	10.677
Wild Boot. p-val					0.497		0.029
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the standardized Coping Strategies Index (CSI) multiplied by -1; thus, a negative coefficient represents an increase in coping intensity (lower food security) and a positive coefficient represents a decrease in coping intensity (higher food security). The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.9
Refugee Households: Effect of Refugee Arrivals on Food Security

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.0063 (0.0193)	-0.0270 (0.0215)	0.0046 (0.0277)	-0.0017 (0.0275)	-0.0017 (0.0698)	0.1742*** (0.0661)	0.1742 (0.1531)
Observations	2990	2990	2990	2990	2990	2965	2965
R-squared	0.073	0.080	0.094	0.071	0.071	0.051	0.051
UnderID (LM Test)						347.937	3.760
UnderID P-Value						0.000	0.052
Weak ID (Wald F-Stat)						422.383	19.134
Wild Boot. p-val					0.978		0.810
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	0.0137** (0.0061)	0.0029 (0.0081)	0.0096 (0.0104)	0.0058 (0.0101)	0.0058 (0.0171)	-0.0899*** (0.0313)	-0.0899** (0.0497)
Observations	2990	2990	2990	2990	2990	2965	2965
R-squared	0.074	0.080	0.095	0.071	0.071	0.030	0.030
UnderID (LM Test)						223.373	5.599
UnderID P-Value						0.000	0.018
Weak ID (Wald F-Stat)						251.235	6.804
Wild Boot. p-val					0.798		0.024
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	0.0087*** (0.0026)	0.0025 (0.0037)	0.0147*** (0.0049)	0.0120** (0.0047)	0.0120 (0.0081)	-0.0477*** (0.0164)	-0.0477** (0.0238)
Observations	2990	2990	2990	2990	2990	2965	2965
R-squared	0.076	0.080	0.097	0.073	0.073	0.004	0.004
UnderID (LM Test)						191.115	6.116
UnderID P-Value						0.000	0.013
Weak ID (Wald F-Stat)						210.938	11.967
Wild Boot. p-val					0.209		0.012
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the standardized Coping Strategies Index (CSI) multiplied by -1; thus, a negative coefficient represents an increase in coping intensity (lower food security) and a positive coefficient represents a decrease in coping intensity (higher food security). The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.10
Host Households: Effect of Refugee Arrivals on Any Harvest Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	−0.0021 (0.0038)	−0.0047 (0.0040)	0.0032 (0.0051)	0.0032 (0.0050)	0.0032 (0.0082)	0.0190** (0.0114)	0.0190* (0.0118)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.027	0.028	0.044	0.024	0.024	0.020	0.020
UnderID (LM Test)						416.324	4.343
UnderID P-Value						0.000	0.037
Weak ID (Wald F-Stat)						511.249	9.434
Wild Boot. p-val					0.744		0.045
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	−0.0013 (0.0014)	−0.0034** (0.0015)	−0.0028 (0.0020)	−0.0029 (0.0019)	−0.0029 (0.0017)	0.0230*** (0.0065)	0.0230 (0.0184)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.027	0.029	0.045	0.025	0.025	−0.056	−0.056
UnderID (LM Test)						209.246	5.910
UnderID P-Value						0.000	0.015
Weak ID (Wald F-Stat)						230.147	5.798
Wild Boot. p-val					0.415		0.228
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	−0.0004 (0.0005)	−0.0013** (0.0006)	−0.0011 (0.0010)	−0.0012 (0.0009)	−0.0012 (0.0010)	0.0102*** (0.0030)	0.0102 (0.0075)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.027	0.028	0.044	0.025	0.025	−0.041	−0.041
UnderID (LM Test)						221.856	6.472
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						245.576	10.680
Wild Boot. p-val					0.509		0.243
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary indicator for having any harvest (non-zero kg harvested). The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.11
Refugee Households: Effect of Refugee Arrivals on Any Harvest Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	−0.0442*** (0.0082)	−0.0442*** (0.0082)	−0.0246*** (0.0092)	−0.0216** (0.0089)	−0.0216 (0.0130)	0.0060 (0.0214)	0.0060 (0.0346)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.081	0.127	0.159	0.149	0.149	0.146	0.146
UnderID (LM Test)						362.461	3.645
UnderID P-Value						0.000	0.056
Weak ID (Wald F-Stat)						441.729	18.466
Wild Boot. p-val					0.295		0.946
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	−0.0197*** (0.0029)	−0.0231*** (0.0033)	−0.0142*** (0.0038)	−0.0134*** (0.0036)	−0.0134** (0.0045)	0.0227** (0.0102)	0.0227 (0.0245)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.092	0.137	0.162	0.154	0.154	0.106	0.106
UnderID (LM Test)						233.808	5.385
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						263.764	7.512
Wild Boot. p-val					0.118		0.580
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	−0.0090*** (0.0011)	−0.0081*** (0.0014)	−0.0036** (0.0018)	−0.0035** (0.0017)	−0.0035 (0.0027)	0.0104** (0.0050)	0.0104 (0.0111)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.098	0.130	0.158	0.149	0.149	0.118	0.118
UnderID (LM Test)						220.949	5.605
UnderID P-Value						0.000	0.018
Weak ID (Wald F-Stat)						247.421	16.062
Wild Boot. p-val					0.470		0.581
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary indicator for having any harvest (non-zero kg harvested). The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.12
Host Households: Effect of Refugee Arrivals on Total Harvest (Log KG)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.1306*** (0.0318)	-0.1370*** (0.0346)	-0.0386 (0.0439)	-0.0548 (0.0403)	-0.0548 (0.0880)	0.0562 (0.0966)	0.0562 (0.1136)
Observations	3254	3254	3254	3254	3254	3226	3226
R-squared	0.064	0.069	0.100	0.069	0.069	0.066	0.066
UnderID (LM Test)						394.586	4.286
UnderID P-Value						0.000	0.038
Weak ID (Wald F-Stat)						484.337	9.233
Wild Boot. p-val					0.890		0.648
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0560*** (0.0109)	-0.0838*** (0.0126)	-0.0601*** (0.0166)	-0.0591*** (0.0154)	-0.0591*** (0.0176)	0.1683*** (0.0555)	0.1683 (0.1446)
Observations	3254	3254	3254	3254	3254	3226	3226
R-squared	0.068	0.076	0.103	0.074	0.074	-0.014	-0.014
UnderID (LM Test)						193.744	5.789
UnderID P-Value						0.000	0.016
Weak ID (Wald F-Stat)						212.504	5.606
Wild Boot. p-val					0.266		0.228
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0136*** (0.0043)	-0.0276*** (0.0053)	-0.0304*** (0.0082)	-0.0274*** (0.0075)	-0.0274** (0.0110)	0.0743*** (0.0253)	0.0743 (0.0573)
Observations	3254	3254	3254	3254	3254	3226	3226
R-squared	0.063	0.072	0.103	0.073	0.073	-0.001	-0.001
UnderID (LM Test)						217.685	6.395
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						241.831	10.404
Wild Boot. p-val					0.203		0.234
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total harvest (kg) in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.13
Refugee Households: Effect of Refugee Arrivals on Total Harvest (Log KG)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.2735*** (0.0427)	-0.2671*** (0.0452)	-0.0061 (0.0524)	0.0216 (0.0492)	0.0216 (0.0875)	0.2709** (0.1263)	0.2709 (0.2936)
Observations	2834	2834	2834	2834	2834	2793	2793
R-squared	0.132	0.173	0.274	0.057	0.057	0.043	0.043
UnderID (LM Test)						276.777	4.177
UnderID P-Value						0.000	0.041
Weak ID (Wald F-Stat)						326.120	20.570
Wild Boot. p-val					0.830		0.504
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.1397*** (0.0136)	-0.1619*** (0.0163)	-0.0555*** (0.0202)	-0.0390** (0.0187)	-0.0390 (0.0289)	0.1387** (0.0544)	0.1387 (0.1213)
Observations	2834	2834	2834	2834	2834	2793	2793
R-squared	0.154	0.193	0.276	0.059	0.059	0.009	0.009
UnderID (LM Test)						209.920	5.513
UnderID P-Value						0.000	0.019
Weak ID (Wald F-Stat)						236.688	7.857
Wild Boot. p-val					0.349		0.461
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0736*** (0.0053)	-0.0814*** (0.0068)	-0.0178* (0.0092)	-0.0139 (0.0086)	-0.0139 (0.0153)	0.0597** (0.0257)	0.0597 (0.0551)
Observations	2834	2834	2834	2834	2834	2793	2793
R-squared	0.178	0.204	0.275	0.058	0.058	0.018	0.018
UnderID (LM Test)						200.623	5.389
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						224.859	17.024
Wild Boot. p-val					0.554		0.547
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total harvest (kg) in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.14
Host Households: Effect of Refugee Arrivals on Any Earnings Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	0.0244*** (0.0083)	0.0187** (0.0091)	0.0315*** (0.0121)	0.0294** (0.0118)	0.0294** (0.0109)	0.0123 (0.0278)	0.0123 (0.0320)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.049	0.054	0.058	0.018	0.018	0.017	0.017
UnderID (LM Test)						416.324	4.343
UnderID P-Value						0.000	0.037
Weak ID (Wald F-Stat)						511.249	9.434
Wild Boot. p-val					0.013		0.793
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	0.0050* (0.0027)	0.0033 (0.0034)	0.0060 (0.0046)	0.0061 (0.0043)	0.0061 (0.0042)	0.0642*** (0.0157)	0.0642** (0.0375)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.048	0.053	0.057	0.016	0.016	-0.053	-0.053
UnderID (LM Test)						209.070	5.793
UnderID P-Value						0.000	0.015
Weak ID (Wald F-Stat)						229.653	5.691
Wild Boot. p-val					0.232		0.048
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	0.0011 (0.0011)	0.0018 (0.0014)	0.0007 (0.0023)	0.0010 (0.0021)	0.0010 (0.0017)	0.0293*** (0.0074)	0.0293* (0.0151)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.047	0.053	0.056	0.015	0.015	-0.053	-0.053
UnderID (LM Test)						221.856	6.472
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						245.576	10.680
Wild Boot. p-val					0.551		0.081
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary indicator for having any agricultural earnings in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.15
Refugee Households: Effect of Refugee Arrivals on Any Earnings Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	−0.0380*** (0.0079)	−0.0416*** (0.0089)	−0.0157 (0.0116)	−0.0167 (0.0113)	−0.0167 (0.0277)	−0.0886*** (0.0274)	−0.0886 (0.0553)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.040	0.059	0.115	0.059	0.059	0.040	0.040
UnderID (LM Test)						362.461	3.645
UnderID P-Value						0.000	0.056
Weak ID (Wald F-Stat)						441.729	18.466
Wild Boot. p-val					0.640		0.426
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	−0.0167*** (0.0024)	−0.0224*** (0.0031)	−0.0073* (0.0042)	−0.0083** (0.0040)	−0.0083 (0.0060)	0.0225* (0.0128)	0.0225* (0.0150)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.045	0.065	0.116	0.059	0.059	0.036	0.036
UnderID (LM Test)						233.808	5.385
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						263.764	7.512
Wild Boot. p-val					0.338		0.073
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	−0.0085*** (0.0011)	−0.0100*** (0.0014)	0.0012 (0.0020)	0.0009 (0.0018)	0.0009 (0.0028)	0.0122* (0.0062)	0.0122* (0.0072)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.051	0.066	0.115	0.058	0.058	0.044	0.044
UnderID (LM Test)						220.949	5.605
UnderID P-Value						0.000	0.018
Weak ID (Wald F-Stat)						247.421	16.062
Wild Boot. p-val					0.820		0.087
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary indicator for having any agricultural earnings in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.16
Host Households: Effect of Refugee Arrivals on Agricultural Earnings (Log USD)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	0.0822** (0.0405)	0.0570 (0.0450)	0.1294** (0.0598)	0.1376** (0.0559)	0.1376* (0.0686)	0.0921 (0.1310)	0.0921 (0.1462)
Observations	3243	3243	3243	3243	3243	3211	3211
R-squared	0.067	0.069	0.073	0.025	0.025	0.025	0.025
UnderID (LM Test)						402.635	4.304
UnderID P-Value						0.000	0.038
Weak ID (Wald F-Stat)						497.151	9.572
Wild Boot. p-val					0.060		0.639
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	0.0103 (0.0134)	-0.0045 (0.0168)	0.0094 (0.0230)	0.0166 (0.0209)	0.0166 (0.0190)	0.2304*** (0.0749)	0.2304 (0.1824)
Observations	3243	3243	3243	3243	3243	3211	3211
R-squared	0.066	0.068	0.072	0.023	0.023	-0.021	-0.021
UnderID (LM Test)						192.662	5.814
UnderID P-Value						0.000	0.016
Weak ID (Wald F-Stat)						211.295	5.642
Wild Boot. p-val					0.331		0.222
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	0.0032 (0.0054)	0.0018 (0.0072)	-0.0084 (0.0113)	-0.0032 (0.0104)	-0.0032 (0.0126)	0.1031*** (0.0350)	0.1031 (0.0756)
Observations	3243	3243	3243	3243	3243	3211	3211
R-squared	0.066	0.068	0.072	0.022	0.022	-0.023	-0.023
UnderID (LM Test)						210.667	6.430
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						233.277	10.323
Wild Boot. p-val					0.845		0.235
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total agricultural earnings (USD) in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.17
Refugee Households: Effect of Refugee Arrivals on Agricultural Earnings (Log USD)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	−0.0935*** (0.0252)	−0.1062*** (0.0299)	0.1005*** (0.0389)	0.1073*** (0.0348)	0.1073* (0.0574)	−0.0764 (0.0893)	−0.0764 (0.1545)
Observations	2838	2838	2838	2838	2838	2799	2799
R-squared	0.066	0.075	0.188	0.019	0.019	0.004	0.004
UnderID (LM Test)						276.678	4.171
UnderID P-Value						0.000	0.041
Weak ID (Wald F-Stat)						325.849	20.335
Wild Boot. p-val					0.141		0.708
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	−0.0534*** (0.0074)	−0.0757*** (0.0102)	0.0182 (0.0141)	0.0191 (0.0125)	0.0191 (0.0178)	0.0618 (0.0376)	0.0618 (0.0396)
Observations	2838	2838	2838	2838	2838	2799	2799
R-squared	0.074	0.086	0.186	0.015	0.015	0.009	0.009
UnderID (LM Test)						210.323	5.537
UnderID P-Value						0.000	0.019
Weak ID (Wald F-Stat)						237.136	7.984
Wild Boot. p-val					0.566		0.229
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	−0.0294*** (0.0034)	−0.0413*** (0.0046)	0.0113* (0.0067)	0.0111* (0.0058)	0.0111 (0.0086)	0.0311* (0.0178)	0.0311 (0.0190)
Observations	2838	2838	2838	2838	2838	2799	2799
R-squared	0.083	0.095	0.187	0.016	0.016	0.009	0.009
UnderID (LM Test)						202.210	5.387
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						226.806	17.240
Wild Boot. p-val					0.384		0.106
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total agricultural earnings (USD) in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.18
Host Households: Effect of Refugee Arrivals on Total Kilograms Sold (Log)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	0.0525 (0.0474)	0.0209 (0.0531)	0.1311* (0.0716)	0.1232* (0.0669)	0.1232 (0.0798)	0.0577 (0.1568)	0.0577 (0.1995)
Observations	3254	3254	3254	3254	3254	3228	3228
R-squared	0.082	0.087	0.093	0.023	0.023	0.023	0.023
UnderID (LM Test)						401.149	4.333
UnderID P-Value						0.000	0.037
Weak ID (Wald F-Stat)						494.317	9.526
Wild Boot. p-val					0.139		0.819
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0091 (0.0155)	-0.0297 (0.0196)	0.0030 (0.0274)	0.0066 (0.0248)	0.0066 (0.0246)	0.3148*** (0.0897)	0.3148 (0.2268)
Observations	3254	3254	3254	3254	3254	3228	3228
R-squared	0.082	0.087	0.092	0.022	0.022	-0.041	-0.041
UnderID (LM Test)						195.742	5.795
UnderID P-Value						0.000	0.016
Weak ID (Wald F-Stat)						214.923	5.725
Wild Boot. p-val					0.814		0.195
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0070 (0.0063)	-0.0086 (0.0084)	-0.0109 (0.0135)	-0.0069 (0.0123)	-0.0069 (0.0136)	0.1392*** (0.0409)	0.1392 (0.0898)
Observations	3254	3254	3254	3254	3254	3228	3228
R-squared	0.082	0.087	0.093	0.022	0.022	-0.038	-0.038
UnderID (LM Test)						221.857	6.403
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						247.019	10.934
Wild Boot. p-val					0.654		0.218
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total kilograms sold in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.19
Refugee Households: Effect of Refugee Arrivals on Total Kilograms Sold (Log)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV (Clust.)	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.1226*** (0.0350)	-0.1383*** (0.0417)	0.1470*** (0.0537)	0.1576*** (0.0494)	0.1576 (0.0918)	-0.0806 (0.1213)	-0.0806 (0.2283)
Observations	2833	2833	2833	2833	2833	2793	2793
R-squared	0.073	0.083	0.204	0.020	0.020	0.006	0.006
UnderID (LM Test)						276.712	4.181
UnderID P-Value						0.000	0.041
Weak ID (Wald F-Stat)						326.062	20.565
Wild Boot. p-val					0.187		0.773
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0766*** (0.0103)	-0.1075*** (0.0141)	0.0233 (0.0192)	0.0244 (0.0175)	0.0244 (0.0229)	0.0623 (0.0514)	0.0623 (0.0648)
Observations	2833	2833	2833	2833	2833	2793	2793
R-squared	0.082	0.096	0.202	0.015	0.015	0.012	0.012
UnderID (LM Test)						207.999	5.523
UnderID P-Value						0.000	0.019
Weak ID (Wald F-Stat)						234.250	7.861
Wild Boot. p-val					0.589		0.456
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0435*** (0.0046)	-0.0606*** (0.0063)	0.0152* (0.0092)	0.0146* (0.0082)	0.0146 (0.0109)	0.0325 (0.0244)	0.0325 (0.0293)
Observations	2833	2833	2833	2833	2833	2793	2793
R-squared	0.094	0.106	0.202	0.015	0.015	0.013	0.013
UnderID (LM Test)						198.637	5.381
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						222.366	17.034
Wild Boot. p-val					0.389		0.342
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes ^a	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the log of total kilograms sold in the past year. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.20
Host Households: Effect of Refugee Arrivals on Any Livestock Ownership Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	0.0050 (0.0084)	0.0148 (0.0095)	0.0414*** (0.0120)	0.0459*** (0.0113)	0.0459* (0.0216)	−0.0300 (0.0252)	−0.0300 (0.0371)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.116	0.131	0.200	0.164	0.164	0.145	0.145
UnderID (LM Test)						416.324	4.343
UnderID P-Value						0.000	0.037
Weak ID (Wald F-Stat)						511.249	9.434
Wild Boot. p-val					0.335		0.359
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	−0.0048* (0.0029)	0.0010 (0.0037)	0.0123** (0.0049)	0.0131*** (0.0044)	0.0131** (0.0046)	0.0569*** (0.0140)	0.0569** (0.0237)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.117	0.130	0.199	0.161	0.161	0.120	0.120
UnderID (LM Test)						209.246	5.910
UnderID P-Value						0.000	0.015
Weak ID (Wald F-Stat)						230.147	5.798
Wild Boot. p-val					0.487		0.000
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	−0.0029*** (0.0011)	0.0013 (0.0015)	0.0026 (0.0024)	0.0028 (0.0021)	0.0028 (0.0050)	0.0265*** (0.0067)	0.0265*** (0.0102)
Observations	3367	3367	3367	3367	3367	3357	3357
R-squared	0.118	0.130	0.198	0.158	0.158	0.107	0.107
UnderID (LM Test)						221.856	6.472
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						245.576	10.680
Wild Boot. p-val					0.709		0.000
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary variable indicating whether the household owns any livestock. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.21
Refugee Households: Effect of Refugee Arrivals on Any Livestock Ownership Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.0238*** (0.0086)	-0.0264*** (0.0081)	-0.0065 (0.0103)	-0.0075 (0.0102)	-0.0075 (0.0303)	0.0855*** (0.0251)	0.0855 (0.0832)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.217	0.360	0.406	0.483	0.483	0.463	0.463
UnderID (LM Test)						362.461	3.645
UnderID P-Value						0.000	0.056
Weak ID (Wald F-Stat)						441.729	18.466
Wild Boot. p-val					0.868		0.480
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0173*** (0.0027)	-0.0197*** (0.0032)	-0.0177*** (0.0039)	-0.0179*** (0.0037)	-0.0179** (0.0072)	-0.0002 (0.0115)	-0.0002 (0.0220)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.225	0.366	0.410	0.488	0.488	0.483	0.483
UnderID (LM Test)						233.808	5.385
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						263.764	7.512
Wild Boot. p-val					0.225		0.993
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0142*** (0.0011)	-0.0110*** (0.0014)	-0.0061*** (0.0019)	-0.0061*** (0.0018)	-0.0061 (0.0038)	-0.0017 (0.0056)	-0.0017 (0.0107)
Observations	3037	3037	3037	3037	3037	3020	3020
R-squared	0.254	0.371	0.408	0.486	0.486	0.484	0.484
UnderID (LM Test)						220.949	5.605
UnderID P-Value						0.000	0.018
Weak ID (Wald F-Stat)						247.421	16.062
Wild Boot. p-val					0.335		0.909
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is a binary variable indicating whether the refugee household owns any livestock. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.22
Host Households: Effect of Refugee Arrivals on Total Livestock Heads (Log)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.0389** (0.0161)	-0.0085 (0.0184)	0.0475** (0.0232)	0.0597*** (0.0228)	0.0597 (0.0487)	-0.1482*** (0.0566)	-0.1482* (0.0869)
Observations	3312	3312	3312	3312	3312	3293	3293
R-squared	0.139	0.145	0.222	0.130	0.130	0.100	0.100
UnderID (LM Test)						398.352	4.280
UnderID P-Value						0.000	0.039
Weak ID (Wald F-Stat)						486.947	9.137
Wild Boot. p-val					0.588		0.049
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0208*** (0.0051)	-0.0066 (0.0065)	0.0148* (0.0090)	0.0165** (0.0084)	0.0165 (0.0140)	0.1577*** (0.0317)	0.1577** (0.0637)
Observations	3312	3312	3312	3312	3312	3293	3293
R-squared	0.141	0.145	0.222	0.129	0.129	0.036	0.036
UnderID (LM Test)						205.540	5.963
UnderID P-Value						0.000	0.015
Weak ID (Wald F-Stat)						226.210	6.002
Wild Boot. p-val					0.785		0.026
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0065*** (0.0022)	0.0021 (0.0029)	0.0027 (0.0045)	0.0025 (0.0042)	0.0025 (0.0110)	0.0737*** (0.0149)	0.0737*** (0.0278)
Observations	3312	3312	3312	3312	3312	3293	3293
R-squared	0.140	0.145	0.221	0.128	0.128	0.028	0.028
UnderID (LM Test)						221.227	6.437
UnderID P-Value						0.000	0.011
Weak ID (Wald F-Stat)						245.465	11.068
Wild Boot. p-val					0.868		0.033
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the natural log of livestock quantity. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.23
Refugee Households: Effect of Refugee Arrivals on Total Livestock Heads (Log)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FE-Year	TWFE-Dist	TWFE-HH	TWFE-HH (Clust.)	TWFE-IV	TWFE-IV (Clust.)
<i>Panel A: 3 Month Lag</i>							
Change in Arrivals	-0.0417*** (0.0154)	-0.0471*** (0.0146)	-0.0199 (0.0192)	-0.0252 (0.0190)	-0.0252 (0.0502)	0.1513*** (0.0480)	0.1513 (0.1434)
Observations	3006	3006	3006	3006	3006	2986	2986
R-squared	0.206	0.342	0.382	0.453	0.453	0.432	0.432
UnderID (LM Test)						356.748	3.655
UnderID P-Value						0.000	0.056
Weak ID (Wald F-Stat)						434.724	18.756
Wild Boot. p-val					0.792		0.463
<i>Panel B: 6 Month Lag</i>							
Change in Arrivals	-0.0290*** (0.0044)	-0.0334*** (0.0053)	-0.0342*** (0.0068)	-0.0355*** (0.0065)	-0.0355** (0.0110)	-0.0108 (0.0219)	-0.0108 (0.0442)
Observations	3006	3006	3006	3006	3006	2986	2986
R-squared	0.212	0.347	0.386	0.459	0.459	0.456	0.456
UnderID (LM Test)						231.432	5.387
UnderID P-Value						0.000	0.020
Weak ID (Wald F-Stat)						261.259	7.637
Wild Boot. p-val					0.179		0.884
<i>Panel C: 12 Month Lag</i>							
Change in Arrivals	-0.0240*** (0.0019)	-0.0176*** (0.0023)	-0.0111*** (0.0032)	-0.0116*** (0.0030)	-0.0116* (0.0055)	-0.0080 (0.0106)	-0.0080 (0.0218)
Observations	3006	3006	3006	3006	3006	2986	2986
R-squared	0.236	0.351	0.384	0.456	0.456	0.456	0.456
UnderID (LM Test)						221.957	5.567
UnderID P-Value						0.000	0.018
Weak ID (Wald F-Stat)						249.177	16.487
Wild Boot. p-val					0.233		0.828
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
District FE	No	No	Yes	No	No	No	No
Household FE	No	No	No	Yes	Yes	Yes	Yes
Clustered SE	No	No	No	No	Yes	No	Yes

Notes: The dependent variable is the natural log of livestock quantity. The independent variable is the change in refugee arrivals (in thousands) over the preceding 3, 6, or 12 months, as indicated. Standard errors are reported in parentheses and are clustered at the district level in columns (5) and (7). For clustered specifications, p-values from a wild cluster bootstrap (using Webb weights) are reported. Anderson-Rubin (AR) tests are performed for F-stats below 16.38. These AR p-values are robust to weak identification. Column (3) includes district fixed effects. Columns (4)-(7) include household fixed effects. Columns (6)-(7) use a shift-share instrument for refugee arrivals. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.24: Household Outcomes and Characteristics by Status and Market Participation

	Host Consumer	Host Producer	Refugee Consumer	Refugee Producer
<i>Panel A: Outcomes</i>				
Coping Strategies Index	18.51	15.39	24.73	27.30
Any Harvest	0.94	0.97	0.85	0.88
Harvest (KG)	595.86	1,188.93	149.19	261.89
Any Agricultural Earnings	0.40	0.76	0.19	0.48
Agricultural Earnings (USD)	45.88	135.39	6.84	19.85
Total Crops Sold (KG)	119.26	410.40	23.88	85.66
Any Livestock Ownership	0.75	0.68	0.46	0.41
Livestock Holdings (Heads)	4.25	3.84	2.11	1.92
<i>Panel B: Household Characteristics</i>				
Years of Schooling	7.81	8.18	8.58	7.84
Household Size	6.62	6.54	6.82	5.86
Total Land Holdings (Acres)	4.48	3.80	0.62	0.51
Normalized Asset Index	0.28	0.63	-0.20	-0.48
Voucher Aid (USD)	7.97	7.40	138.54	200.20
Relief Aid (USD)	2.17	1.18	164.50	86.41
Months in Settlement	.	.	146.65	255.57
Observations	1,580	3,105	2,132	2,135

Notes: The table reports sample means by household status (host vs. refugee) and baseline market participation (consumer vs. producer).

Table B.25: Price Effects of Refugee Arrivals

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: $\ln(\text{Price})$	Beans	Maize	Oil	Millet	Sorghum
$\ln(P_{t-1})$	0.9073*** (0.055 4)	0.9727*** (0.051 3)	0.9896*** (0.029 7)	0.5660*** (0.155 9)	1.0012*** (0.056 9)
$(\Delta)\text{Arrivals (0-3 mo)}$	-0.0013 (0.001 6)	-0.0087** (0.003 8)	0.0024 (0.001 9)	0.0017 (0.001 6)	-0.0058 (0.004 0)
$(\Delta)\text{Arrivals (3-6 mo)}$	-0.0053*** (0.001 5)	0.0064*** (0.002 3)	0.0006 (0.001 1)	-0.0028 (0.001 8)	0.0021 (0.003 0)
$(\Delta)\text{Arrivals (6-12 mo)}$	0.0111*** (0.002 2)	0.0081*** (0.002 1)	-0.0001 (0.001 1)	0.0026 (0.001 7)	0.0080*** (0.002 9)
Max Temp (6m Window)	0.0179 (0.078 3)	0.0201 (0.171 1)	0.1105* (0.058 8)	-0.0393 (0.174 6)	-0.1616* (0.088 6)
Precipitation (6m Window)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000* (0.000 0)	0.0000 (0.000 0)
Max Temp ² (6m Window)	-0.0001 (0.001 3)	0.0001 (0.002 8)	-0.0017* (0.000 9)	0.0010 (0.002 9)	0.0031** (0.001 4)
Precipitation ² (6m Window)	0.0000** (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)
Harvest Season	-0.0356*** (0.009 7)	-0.0485*** (0.014 9)		-0.0023 (0.014 6)	-0.0026 (0.010 3)
Observations	856	834	834	275	781
Markets	32	32	32	13	31
AR(2) p-value	0.375	0.302	0.470	0.621	0.765
Hansen p-value	0.606	0.308	0.204	0.754	0.301
σ	0.122	0.344	0.091	0.117	0.159
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity in USD. All models are estimated using the Arellano-Bond GMM estimator. Standard errors, reported in parentheses, are clustered at the settlement level. The $(\Delta)\text{Arrivals}$ variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period.

Non-refugee settlement markets included: Busia, Fort Portal, Gulu, Hoima, Iganga, Jinja, Kapchorwa, Kasese, Lira, Makaratin, Masaka, Masindi, Mbale, Mbarara, Mukono, Owino, Soroti, Tororo, and Wakiso. Due to a lack of available market price data in these non-hosting regions, millet (Column 4) is estimated strictly on the original 13 refugee-hosting settlements.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.26: Price Effects of Refugee Arrivals, Heterogeneity by Food Preferences

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: $\ln(\text{Price})$	Beans	Maize	Oil	Millet	Sorghum
$\ln(P_{t-1})$	0.9073*** (0.055 4)	0.9727*** (0.051 3)	0.9896*** (0.029 7)	0.4446*** (0.118 2)	0.9556*** (0.062 6)
$(\Delta)\text{Arrivals (0–3 mo)}$	−0.0013 (0.001 6)	−0.0087** (0.003 8)	0.0024 (0.001 9)	0.0071 (0.005 3)	−0.0156 (0.011 7)
$(\Delta)\text{Arrivals (3–6 mo)}$	−0.0053*** (0.001 5)	0.0064*** (0.002 3)	0.0006 (0.001 1)	0.0031 (0.003 6)	−0.0151* (0.007 7)
$(\Delta)\text{Arrivals (6–12 mo)}$	0.0111*** (0.002 2)	0.0081*** (0.002 1)	−0.0001 (0.001 1)	0.0038 (0.003 4)	−0.0122 (0.009 6)
$(\Delta)\text{Arr. (0–3 mo)} \times \text{S. Sudan}$				−0.0058 (0.006 0)	0.0094 (0.012 8)
$(\Delta)\text{Arr. (3–6 mo)} \times \text{S. Sudan}$				−0.0093* (0.004 9)	0.0210** (0.008 2)
$(\Delta)\text{Arr. (6–12 mo)} \times \text{S. Sudan}$				0.0039 (0.004 3)	0.0230** (0.009 5)
Max Temp (6m Window)	0.0179 (0.078 3)	0.0201 (0.171 1)	0.1105* (0.058 8)	−0.1081 (0.179 6)	−0.0642 (0.061 4)
Precipitation (6m Window)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000** (0.000 0)	0.0000 (0.000 0)
Max Temp ² (6m Window)	−0.0001 (0.001 3)	0.0001 (0.002 8)	−0.0017* (0.000 9)	0.0022 (0.003 0)	0.0015 (0.001 0)
Precipitation ² (6m Window)	0.0000** (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)	0.0000 (0.000 0)
Harvest Season	−0.0356*** (0.009 7)	−0.0485*** (0.014 9)		0.0017 (0.014 0)	0.0003 (0.010 2)
S. Sudan Total Effect (0–3 mo) (<i>p-value</i>)				0.0013 (0.553)	−0.0062 (0.194)
S. Sudan Total Effect (3–6 mo) (<i>p-value</i>)				−0.0061 (0.100)	0.0059 (0.152)
S. Sudan Total Effect (6–12 mo) (<i>p-value</i>)				0.0077 (0.007)	0.0109 (0.000)
Observations	856	834	834	275	781
Settlements	32	32	32	13	31
AR(2) <i>p-value</i>	0.375	0.302	0.470	0.719	0.793
Hansen <i>p-value</i>	0.606	0.308	0.204	0.999	0.849
σ	0.122	0.344	0.091	0.118	0.157
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity in USD. All models are estimated using the Arellano-Bond GMM estimator. The variable “S. Sudan” is an indicator for settlements that are primarily South Sudanese refugees. Standard errors, reported in parentheses, are clustered at the settlement level. The $(\Delta)\text{Arrivals}$ variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period. To satisfy the exclusion restrictions and account for the altered error dynamics of the expanded panel, internal instruments are restricted to deeper lags ($t - 3$ to $t - 5$).

Non-refugee settlement markets included: Busia, Fort Portal, Gulu, Hoima, Iganga, Jinja, Kapchorwa, Kasese, Lira, Makaratin, Masaka, Masindi, Mbale, Mbarara, Mukono, Owino, Soroti, Tororo, and Wakiso. Due to a lack of available market price data in these non-hosting regions, millet (Column 4) is estimated on the original 13 refugee-hosting settlements.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Robustness Checks

C.1 Household Results

To ensure the validity of the instrumental variable design and the stability of the small-cluster inference, I conduct two robustness exercises: a temporal placebo test for pre-trends and a leave-one-district-out jackknife estimation. I also discuss concerns related to the COVID-19 outbreak that occurred during the studies time period (D’Errico et al., 2024; Squarcina & Romano, 2024) and discrepancies between the naive and shift-share estimations (Lal et al., 2024).

C.1.1 Temporal Placebo

To ensure the shift-share estimation is not capturing diverging trends from baseline characteristics, I conduct a temporal placebo test (See Table C.1). Using the preferred specification laid out in Section 4, I regress current household outcomes on a one-period future lead of refugee arrivals, instrumented using the forward-lead of the shift-share instrument. Across most outcomes, future arrivals exhibit no predictive power. For the main welfare estimate, food security, I find no evidence of forward arrivals impacting current welfare. I observe significance in host agricultural earnings and marginal significance for both refugee and host earnings indicators ($p < 0.1$), this is slightly larger than the expected false discovery rate across 16 simultaneous placebo tests. This may also be weak evidence of preemptive commercialization due to forward refugee inflows. Largely, these findings provide evidence that household consumption impacts only manifest *after* the refugee shock, considerably reducing concerns of the instrument capturing settlement trends external from refugee arrivals.

Table C.1
Temporal Placebo: Impact of Future Refugee Arrivals on Current Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Food Security	Any Harvest	ln (Harvest)	Any Earnings	ln (Earnings)	ln (KG Sold)	Any L-Stock	ln (L-Stock)
<i>Panel A: Host Households</i>								
Future Arrivals (12m Lead)	0.056 (0.053)	0.016 (0.007)	0.208 (0.072)	0.071** (0.018)	0.410* (0.082)	0.474 (0.105)	0.015 (0.005)	0.067 (0.031)
Mean Dep. Var.	0.01	0.96	6.07	0.63	2.69	3.40	0.64	1.22
Observations	2198	2142	2044	2142	2024	2038	2142	2104
R-squared	0.030	0.053	0.110	0.034	0.046	0.042	0.014	0.032
Weak ID (Wald F)	7.46	7.48	7.39	7.48	7.51	7.42	7.48	7.47
UnderID P-Val	0.033	0.032	0.032	0.032	0.033	0.031	0.032	0.031
Wild Boot. p-val	0.599	0.405	0.277	0.014	0.075	0.111	0.299	0.421
<i>Panel B: Refugee Households</i>								
Future Arrivals (12m Lead)	0.013 (0.028)	-0.001 (0.019)	0.082 (0.101)	0.004 (0.011)	0.050* (0.028)	0.070 (0.063)	-0.009 (0.010)	0.007 (0.017)
Mean Dep. Var.	-0.05	0.78	3.59	0.23	0.70	1.00	0.28	0.42
Observations	1858	1912	1900	1912	1910	1902	1912	1912
R-squared	0.040	0.038	0.063	0.041	0.037	0.037	0.035	0.038
Weak ID (Wald F)	10.56	10.90	10.84	10.90	10.93	10.83	10.90	10.90
UnderID P-Value	0.014	0.015	0.015	0.015	0.015	0.015	0.015	0.015
Wild Boot. p-val	0.764	0.972	0.803	0.842	0.084	0.452	0.429	0.767
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the results of a temporal placebo test estimating the effect of future refugee arrivals (from the next observational period) on current household food security and agricultural production outcomes. Panel A displays estimates for host households, and Panel B for established refugee households. All models include year and household fixed effects, instrumented using the baseline shift-share formulation. Standard errors (in parentheses) are clustered at the district level. Wild cluster bootstrap p-values (Webb weights) are reported to account for the small number of clusters ($N = 11$).

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.1.2 Small-Cluster Inference

A second empirical challenge is that refugee arrivals are clustered at the district level across only 11 districts. While all baseline specifications employ wild cluster bootstrapped standard errors to correct for small sample instability, inference may be fragile if the aggregate results are driven by outlier districts. To test the sensitivity of the results to spatial outliers, I estimate a leave-one-district-out robustness check. I sequentially re-estimate the preferred 12-month IV specification for food security, while dropping one sample district in each iteration.

Table C.2
Leave-One-Out Robustness: Impact of Refugee Arrivals on Food Security

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Drop Adjumani	Drop Arua	Drop Isingiro	Drop Kamwenge	Drop Kikuube	Drop Kiryandongo	Drop Kyegegwa	Drop Lamwo	Drop Moyo	Drop Yumbe
<i>Panel A: Host Households</i>										
Arrivals (12m Lag)	-0.058* (0.029)	-0.050** (0.032)	-0.098** (0.033)	-0.074* (0.023)	-0.035 (0.020)	-0.070* (0.041)	-0.052 (0.021)	-0.064 (0.095)	-0.067** (0.034)	-0.062*** (0.035)
Mean Dep. Var.	-0.01	-0.01	-0.03	0.04	-0.02	-0.01	-0.04	0.02	-0.04	-0.02
Observations	3204	2887	2696	3039	3107	3239	2993	3075	3036	3198
R-squared	-0.050	-0.042	-0.196	-0.052	0.003	-0.074	-0.005	-0.092	-0.076	-0.060
Weak ID (Wald F)	9.65	6.01	6.65	18.51	7.58	4.80	24.52	0.86	6.68	6.26
UnderID P-Val	0.009	0.014	0.156	0.053	0.027	0.041	0.034	0.364	0.011	0.022
Wild Boot. p-val	0.077	0.012	0.013	0.097	0.185	0.064	0.161	0.408	0.021	0.008
<i>Panel B: Refugee Households</i>										
Arrivals (12m Lag)	-0.047*** (0.025)	-0.064*** (0.020)	-0.122*** (0.093)	-0.044 (0.020)	-0.020 (0.021)	-0.067*** (0.045)	-0.044* (0.025)	-0.054* (0.046)	-0.049** (0.026)	-0.032 (0.028)
Mean Dep. Var.	-0.15	-0.16	-0.07	-0.12	-0.12	-0.13	-0.14	-0.11	-0.14	-0.18
Observations	2729	2687	2521	2634	2712	2656	2623	2606	2900	2617
R-squared	0.005	-0.042	-0.362	0.033	0.066	-0.014	0.034	-0.006	0.003	0.048
Weak ID (Wald F)	9.32	8.02	2.07	23.17	3.97	4.27	8.66	2.64	10.62	6.14
UnderID P-Val	0.014	0.026	0.334	0.049	0.084	0.046	0.107	0.178	0.015	0.028
Wild Boot. p-val	0.001	0.000	0.000	0.255	0.488	0.005	0.072	0.094	0.021	0.173
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the results of a leave-one-district-out (jackknife) robustness check to test the sensitivity of the primary food security findings to small-cluster fragility. The dependent variable in all models is the Coping Strategies Index (CSI). Each column iteratively drops one of the sampling districts from the baseline specification. Panel A displays estimates for host households, and Panel B for refugee households. All models include year and household fixed effects, instrumented using the baseline shift-share formulation. Standard errors (in parentheses) are clustered at the district level. Wild cluster bootstrap p-values (Webb weights) are reported to correct for over-rejection in the 10-cluster sub-samples.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results, presented in Table C.2, demonstrate the stability of the initial findings; refugee arrivals consistently worsen household food security for both refugees and hosts. The point estimates for food security remain negative across all specifications and are largely significant, reflecting the generalized effects of arrivals rather than an artifact of a single settlement community. A small proportion of iterations demonstrate a drop in the first-stage Wald F-statistic, though maintaining a negative point estimate. This reflects a mechanical consequence of a loss in first-stage predictive power rather than second-stage instability.

These exercises also address the concern that COVID-19 disruptions may have differentially affected districts in ways correlated with refugee arrivals (Squarcina & Romano, 2024). If pandemic-related disruptions were generating spurious correlations through time varying settlement characteristics in the instrument, future arrivals may predict current outcomes and lead results to be sensitive to specific district exclusions. Neither pattern emerges, suggesting COVID-related confounding is unlikely to drive the main findings.

C.1.3 Sign Reversal and IV Validity

Appendix Tables B.8 through B.23 report the full set of alternative specifications for all main outcomes in increasing specification intensity. Columns (1)-(6) present non-instrumented estimates, which generally indicate attenuated or positive (negative) associations between arrivals and food security (production outcomes). This divergence between OLS and IV estimates could in principle reflect exclusion restriction concerns (Lal et al., 2024), but is fully consistent with the intentional placement of refugee arrivals into districts with greater absorptive capacity (Participants in the Refugee Settlement Study, 2025) — a form of positive selection that attenuates naive estimates and that the IV corrects for. Combined with the best practice standard error adjustments (Davidson & Flachaire, 2008; Roodman et al., 2019; Staiger & Stock, 1997) and robustness exercises described above, the preferred IV estimates likely capture the causal effect of refugee arrivals on food security and production outcomes in these settlement communities. The household results again align with findings of localized price inflation as the likely primary mechanism driving the observed outcomes.

C.2 Price Results

Tables C.3 and C.4 present the full results for the restricted settlement market robustness checks discussed in Section 7.2. Table C.3 reports the baseline difference GMM estimations, including only the 13 settlement markets. Table C.4 extends this to include interactions with the country of origin for the refugee communities in the settlement. Instruments in these expanded specifications are restricted to shallower lags ($t - 2$ to $t - 4$) to satisfy instrument relevance requirements. Across both tables, the primary medium- to long-term localized inflationary effects of refugee arrivals remain stable and statistically significant.

Table C.3: Price Effects of Refugee Arrivals: Settlement Markets Only

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: $\ln(\text{Price})$	Beans	Maize	Oil	Millet	Sorghum
$\ln(P_{t-1})$	0.8929*** (0.1573)	0.9165*** (0.0519)	0.9196*** (0.0666)	0.5887 (0.3744)	0.9037*** (0.1507)
(Δ)Arrivals (0–3 mo)	−0.0034 (0.0032)	−0.0057** (0.0025)	0.0010 (0.0009)	−0.0020 (0.0029)	−0.0021 (0.0033)
(Δ)Arrivals (3–6 mo)	−0.0060 (0.0037)	0.0029 (0.0023)	0.0009 (0.0012)	−0.0016 (0.0014)	−0.0042 (0.0043)
(Δ)Arrivals (6–12 mo)	0.0107*** (0.0030)	0.0060** (0.0023)	0.0005 (0.0011)	0.0027 (0.0018)	0.0034 (0.0030)
Max Temp (6m Window)	−0.1837 (0.2534)	−0.5307** (0.2191)	0.3347*** (0.0899)	−0.1650 (0.1780)	0.0141 (0.2806)
Precipitation (6m Window)	0.0000 (0.0000)	0.0000*** (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Max Temp ² (6m Window)	0.0032 (0.0040)	0.0094** (0.0035)	−0.0053*** (0.0014)	0.0030 (0.0030)	0.0006 (0.0044)
Precipitation ² (6m Window)	0.0000** (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Harvest Season	−0.0247 (0.0229)	−0.0876*** (0.0256)		0.0009 (0.0179)	−0.0330 (0.0193)
Observations	283	283	283	275	265
Settlements	13	13	13	13	13
AR(2) p-value	0.376	0.974	0.573	0.677	0.301
Hansen p-value	0.904	0.744	0.665	0.893	0.605
σ	0.150	0.159	0.071	0.118	0.190
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity in USD. All models are estimated using the Arellano-Bond GMM estimator. Standard errors, reported in parentheses, are clustered at the settlement level. The (Δ)Arrivals variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period. To satisfy instrument relevancy and account for the altered error dynamics of the reduced panel, internal instruments are restricted to shallower lags ($t - 2$ to $t - 4$).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table C.4: Price Effects of Refugee Arrivals, Heterogeneity by Food Preferences

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: ln(Price)	Beans	Maize	Oil	Millet	Sorghum
ln(Price) _{t-1}	0.8836*** (0.1659)	0.9117*** (0.0541)	0.9200*** (0.0691)	0.5675*** (0.1862)	0.7783*** (0.1659)
(Δ)Arrivals (0–3 mo)	-0.0034 (0.0030)	-0.0057** (0.0024)	0.0010 (0.0009)	0.0024 (0.0050)	-0.0084 (0.0050)
(Δ)Arrivals (3–6 mo)	-0.0060 (0.0036)	0.0030 (0.0023)	0.0008 (0.0011)	0.0010 (0.0031)	-0.0160** (0.0061)
(Δ)Arrivals (6–12 mo)	0.0108*** (0.0028)	0.0061** (0.0023)	0.0005 (0.0011)	0.0018 (0.0036)	-0.0133* (0.0068)
(Δ)Arr. (0–3 mo) × S. Sudan				-0.0033 (0.0052)	0.0057 (0.0045)
(Δ)Arr. (3–6 mo) × S. Sudan				-0.0047 (0.0037)	0.0114 (0.0066)
(Δ)Arr. (6–12 mo) × S. Sudan				0.0049 (0.0041)	0.0209** (0.0081)
Max Temp (6m)	-0.1884 (0.2497)	-0.4790** (0.2008)	0.3172*** (0.0921)	-0.1427 (0.1795)	0.2484 (0.2190)
Precipitation (6m)	-0.0000 (0.0000)	0.0000** (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Max Temp ² (6m)	0.0033 (0.0039)	0.0086** (0.0032)	-0.0050*** (0.0014)	0.0027 (0.0030)	-0.0031 (0.0035)
Precipitation ² (6m)	0.0000** (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	0.0000 (0.0000)
Harvest Season	-0.0246 (0.0226)	-0.0862*** (0.0259)		0.0015 (0.0166)	-0.0200 (0.0215)
S. Sudan Total Effect (0–3 mo)				-0.0009	-0.0027
(<i>p</i> -value)				(0.515)	(0.485)
S. Sudan Total Effect (3–6 mo)				-0.0037	-0.0047
(<i>p</i> -value)				(0.041)	(0.335)
S. Sudan Total Effect (6–12 mo)				0.0067	0.0077
(<i>p</i> -value)				(0.001)	(0.040)
Observations	283	283	283	275	265
Settlements	13	13	13	13	13
AR(2) p-value	0.380	0.970	0.577	0.657	0.317
Hansen p-value	0.883	0.685	0.676	1.000	1.000
Idiosyncratic σ (σ_e)	0.150	0.159	0.071	0.117	0.183
Part of Aid Basket	X	X	X		

Notes: The dependent variable is the natural log of the real price for each commodity. All models are estimated using the Arellano-Bond GMM estimator. The variable "S. Sudan" is an indicator for settlements that are primarily South Sudanese refugees. Standard errors, reported in parentheses, are clustered at the settlement level. The (Δ)Arrivals variable is the change of the refugee population in a given settlement, measured in thousands, over the respective time period. To satisfy instrument relevancy and account for the altered error dynamics of the reduced panel, internal instruments are restricted to shallower lags ($t - 2$ to $t - 4$).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$